

The Liver & biliary system Pancreas & Spleen

By
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28.9.2026

The Liver

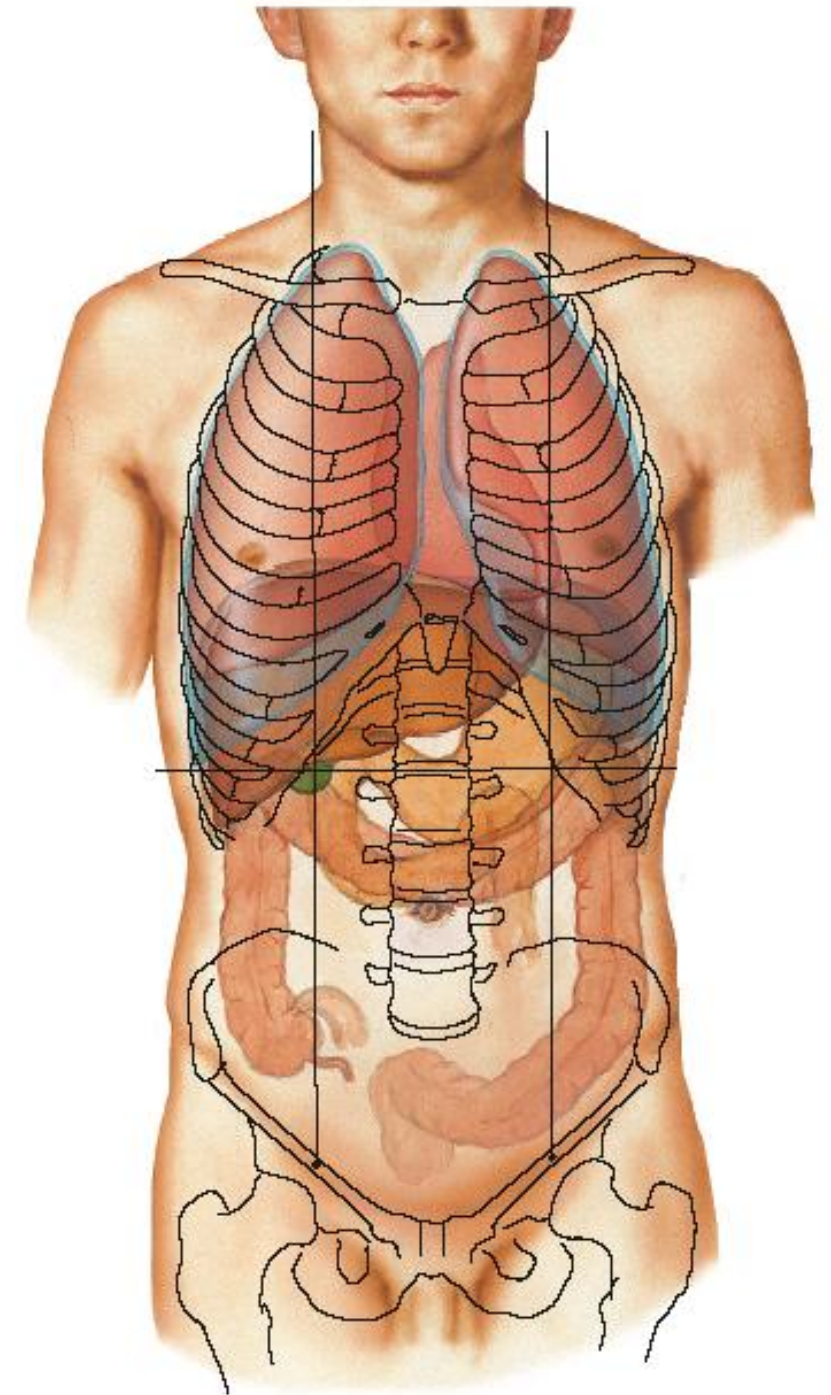
- What is it ? It's the largest gland of the body
- Completely surrounded by a capsule
- Partially surrounded by peritoneum

Functions:

1. Bile secretion
2. CHO, fat, and protein metabolism
3. Heparin formation
4. Remove waste
5. Detoxifying toxins

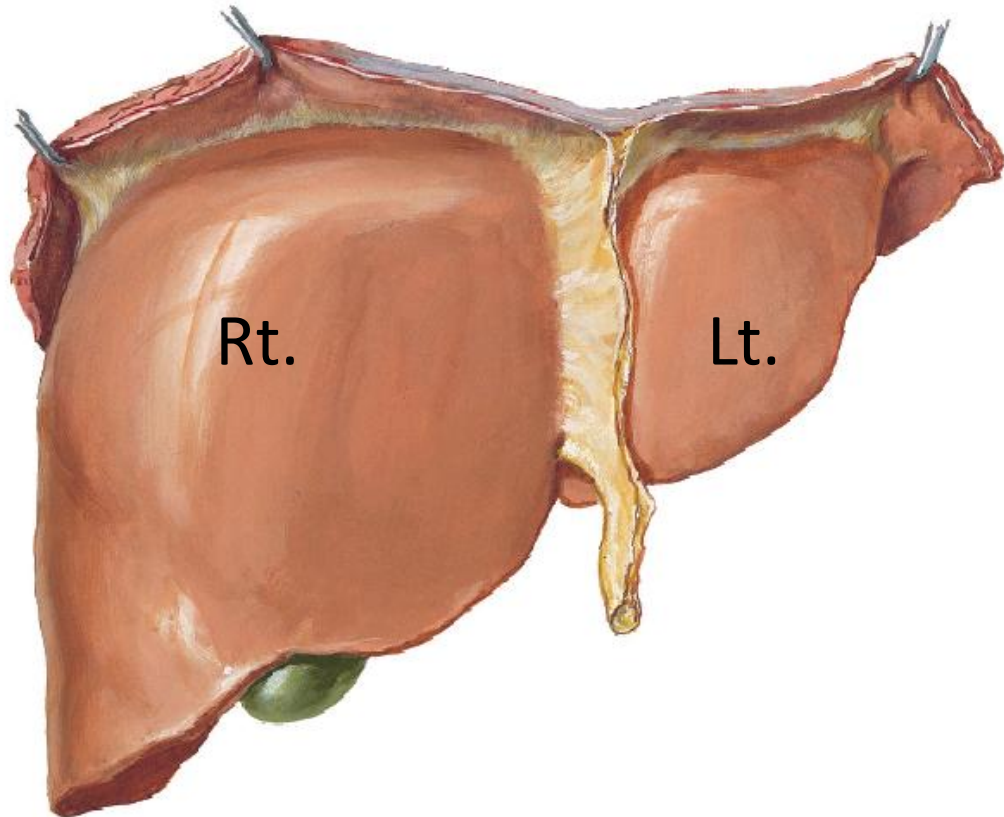
Location :

- Rt upper abdomen
- Beneath diaphragm
- Under cover of Rt. Costal margin
- Under cover of Rt. Hemi diaphragm, which separates it from :
 1. Pleura
 2. Lungs
 3. Pericardium
 4. Heart
- Extends to under cover of Lt. hemi diaphragm

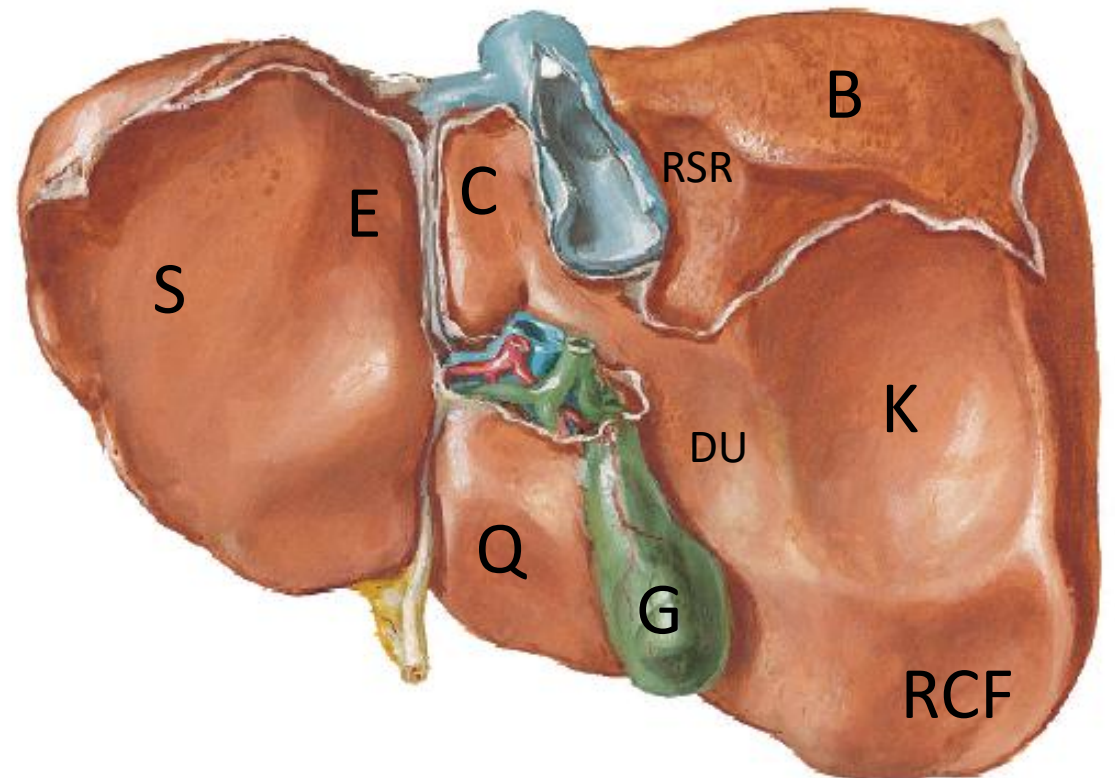


SURFACES :

1. Antero-superior Surface

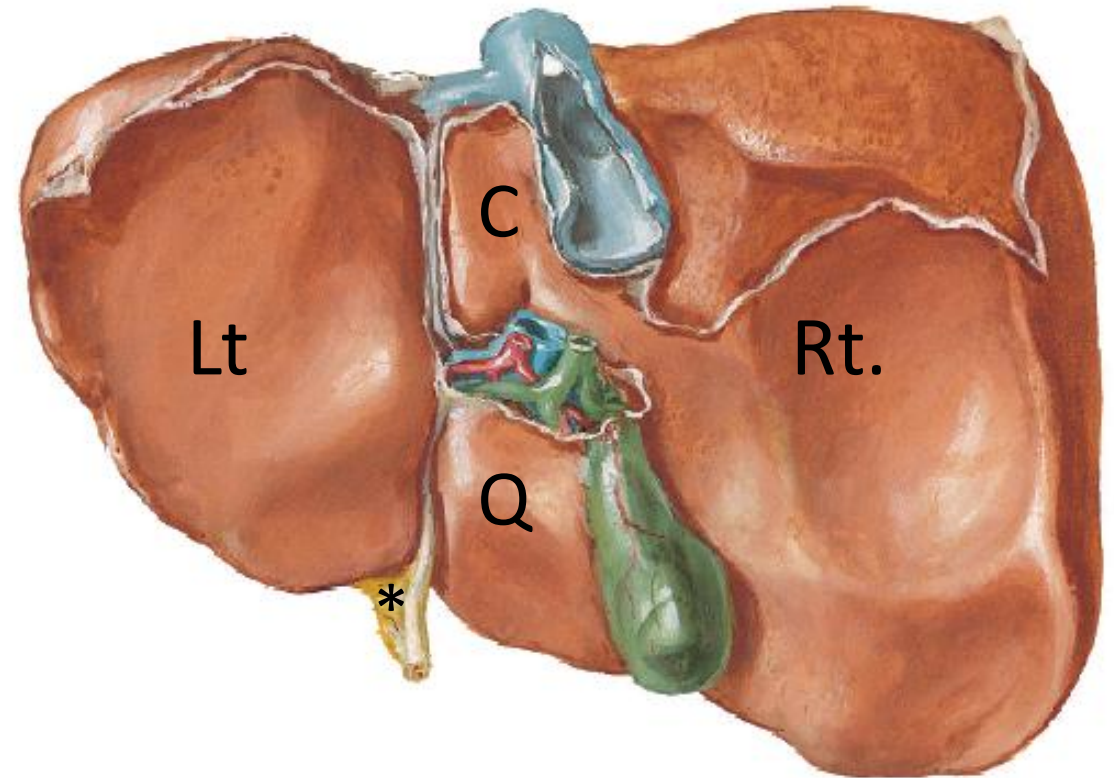


2. Postero-inferior Surface



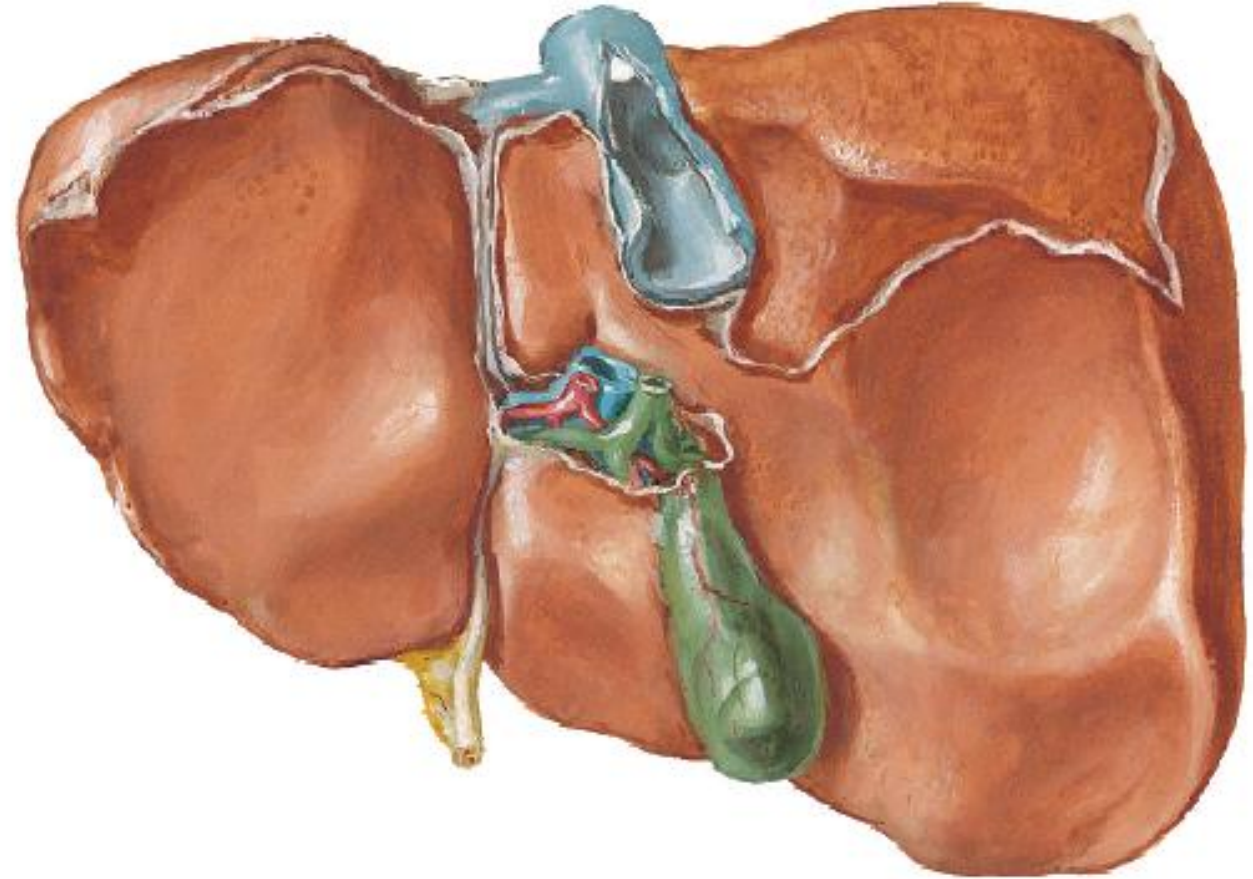
LOBES OF LIVER

- By falciform ligament * its divided to ·
 1. Right Lobe further subdivided to :
 - Caudate Lobe (C)
 - Quadrate Lobe (Q)
 2. Left Lobe



PORTA HEPATIS(HILUM OF LIVER)

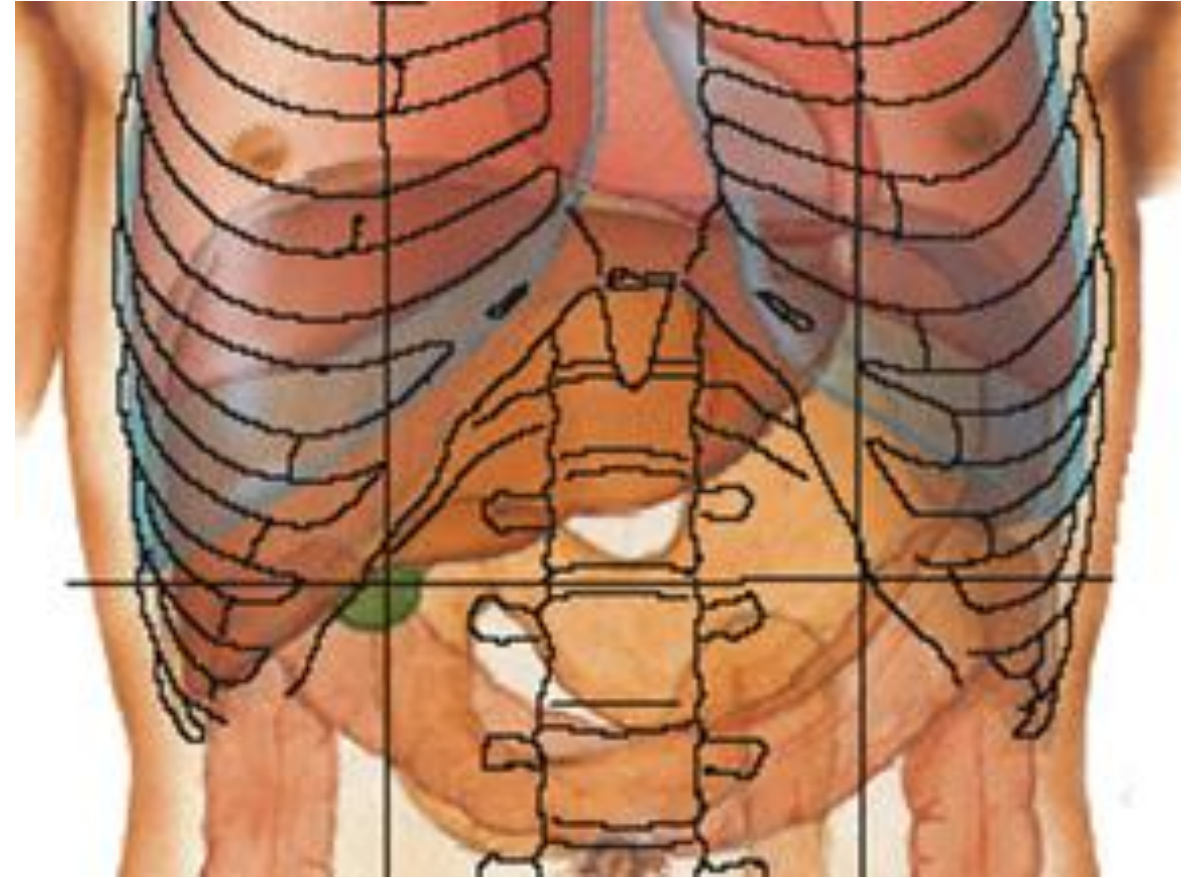
- At the visceral surface
- Between C and Q
- Upper L. Omentum at margins
- Contents :
 1. Rt. & Lt. Hepatic ducts
 2. Rt. & Lt. branches of Hepatic artery
 3. Rt. & Lt. branches of Portal vein
 4. Sympathetic & Parasympathetics
 5. Hepatic LNs.



RELATIONS:

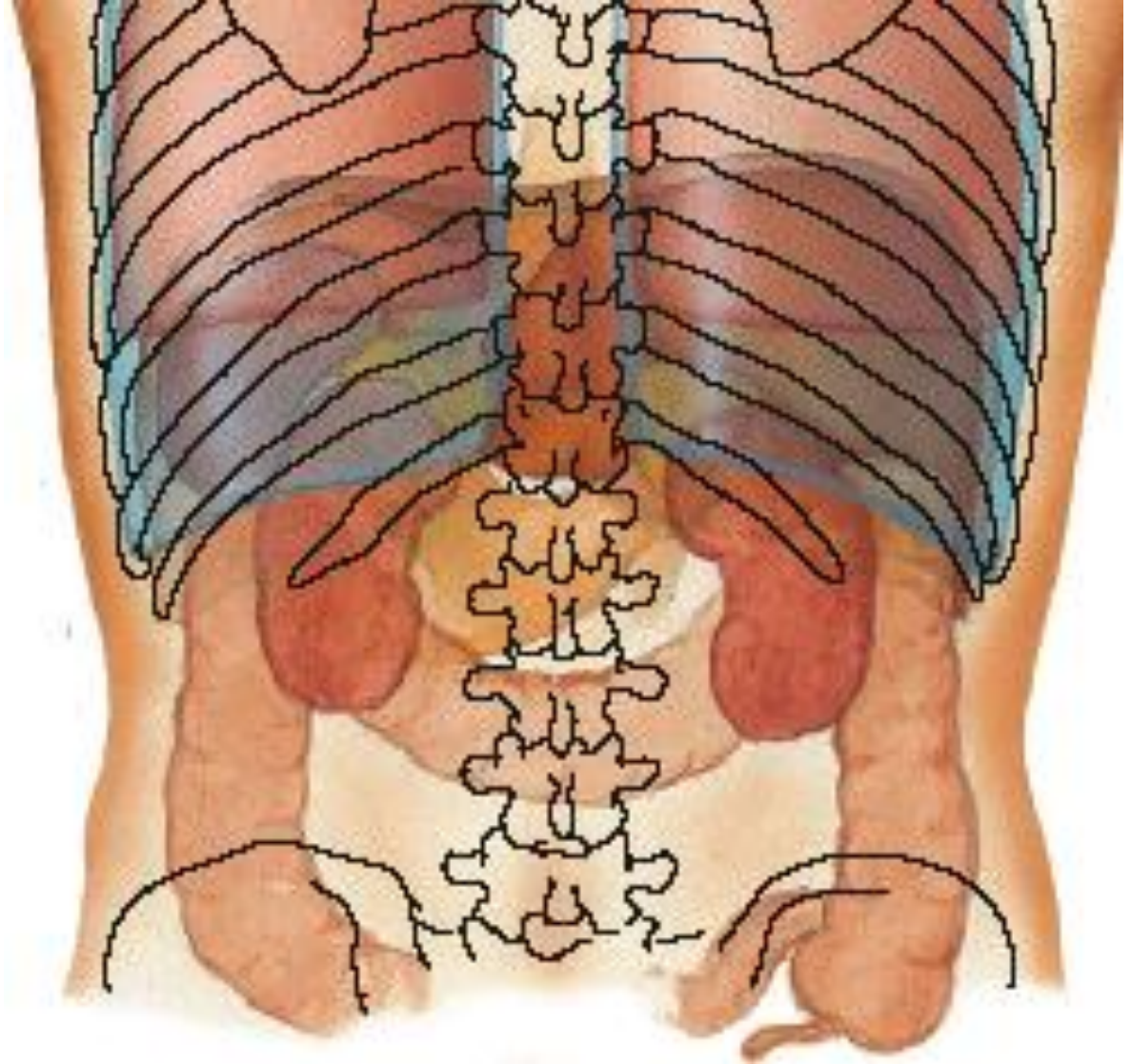
- ANTERIORLY:

1. Diaphragm
2. Rt. & Lt costal margins
3. Rt. & Lt. pleura
4. Lower margins of both lungs
5. Xiphoid process
6. Ant. Abdominal wall in subcostal angle



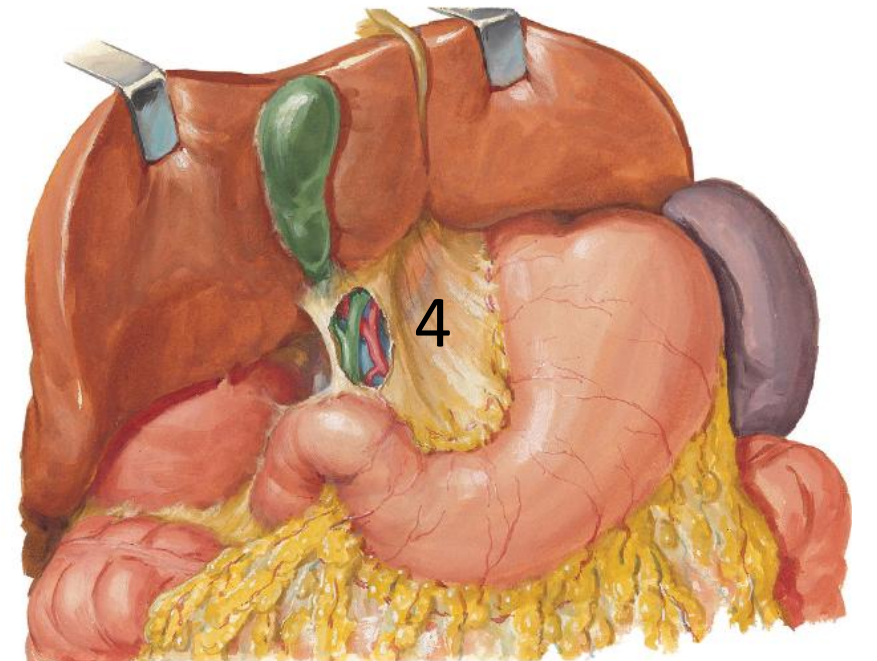
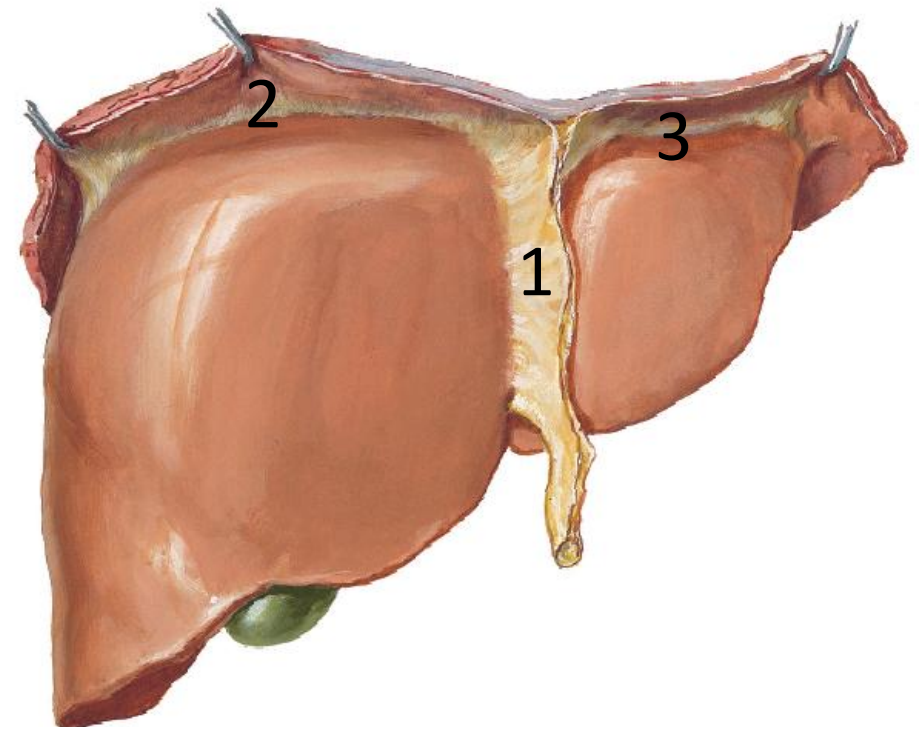
- POSTERIORLY:

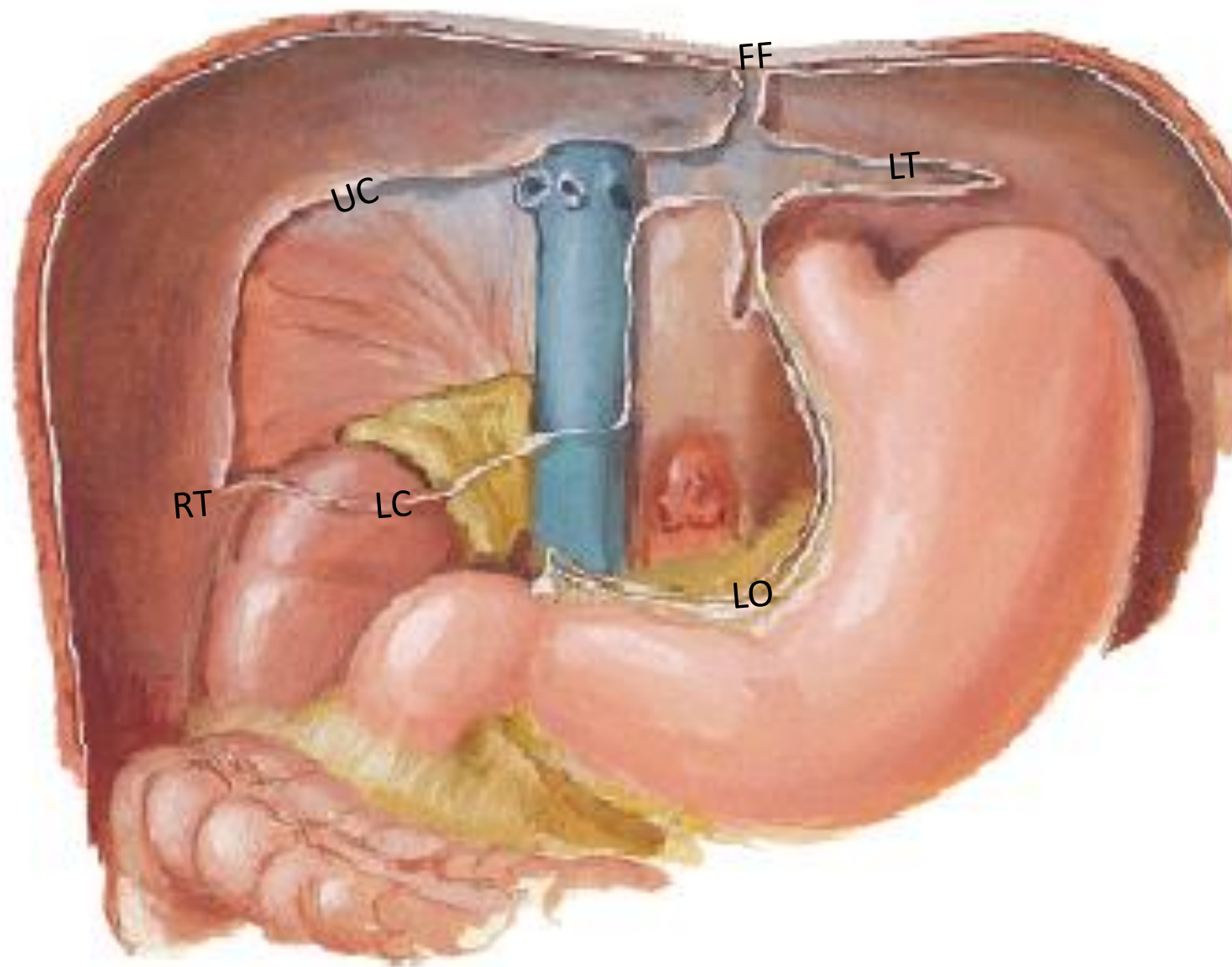
1. Diaphragm & pleura
2. Rt. & Lt. lungs
3. Rt. Kidney
4. Hepatic flexure of colon
5. Duodenum
6. Gallbladder
7. IVC
8. Esophagus
9. Fundus of stomach



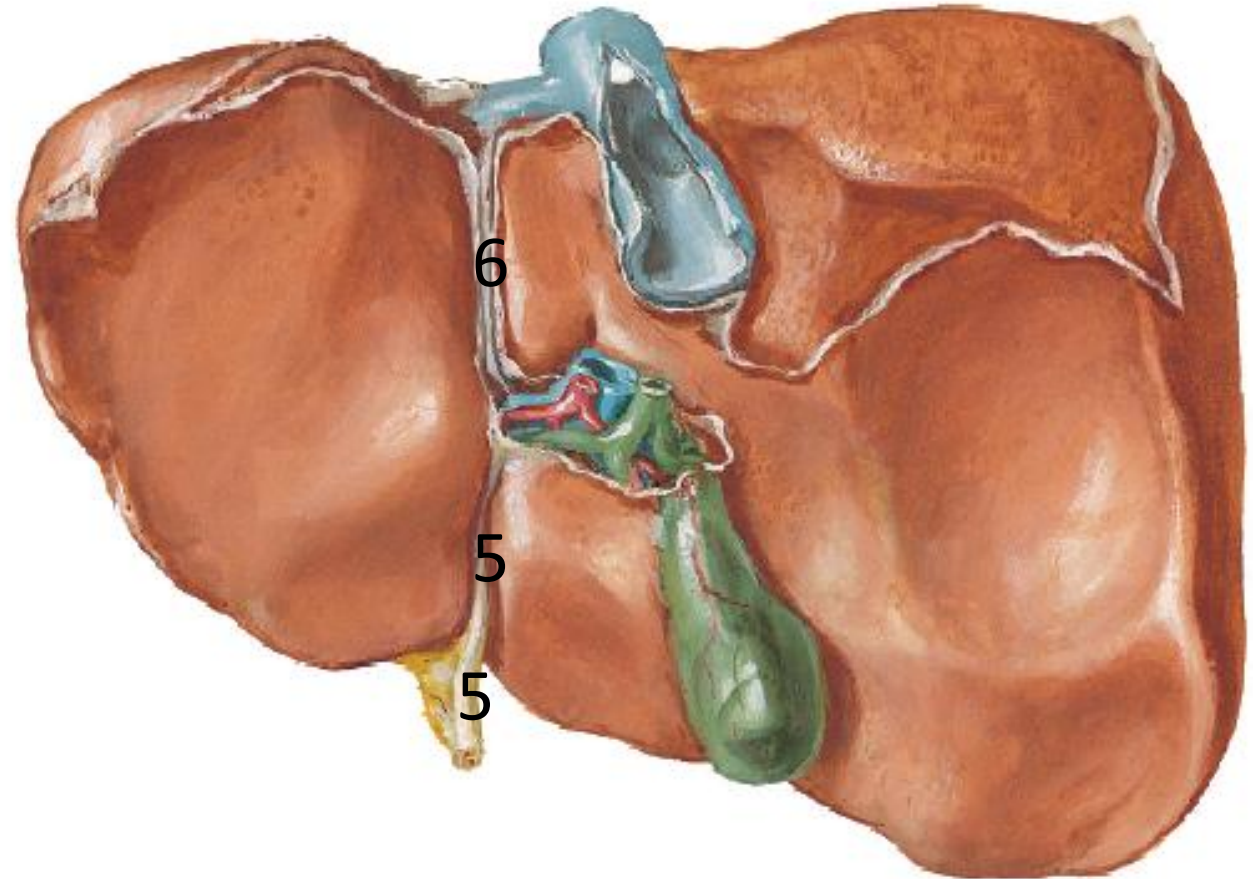
PERITONEAL LIGAMENTS OF LIVER:

1. Falciform ligament
2. Coronary ligament
(upper & lower layers)
3. Triangular ligament
(Right & Left)
4. Lesser omentum

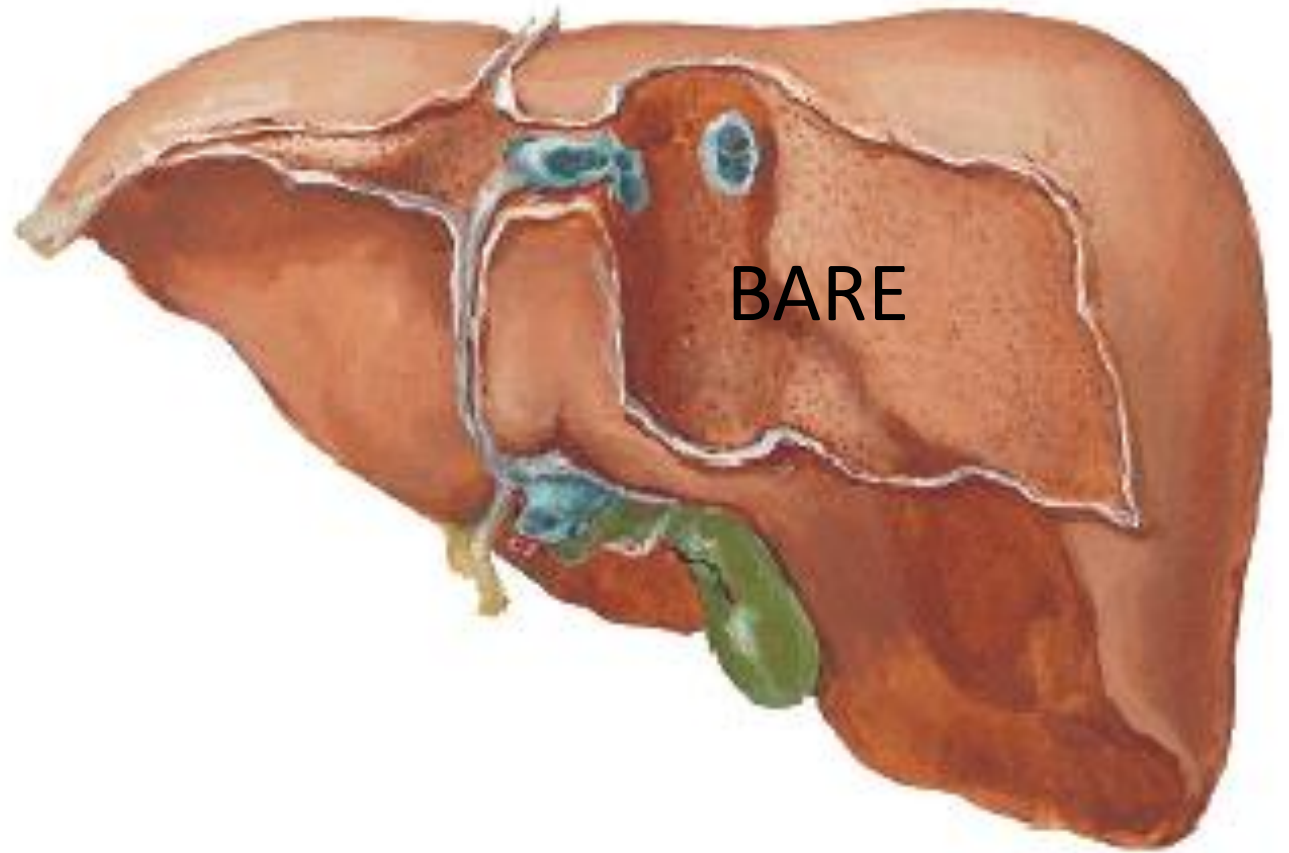




- 5. Ligamentum teres
(Oblit. Lt. paraumb. v.)
- 6. Ligamentum venosum
(Ductus venosus)



- what's Bare area?



1. ARTERIAL :

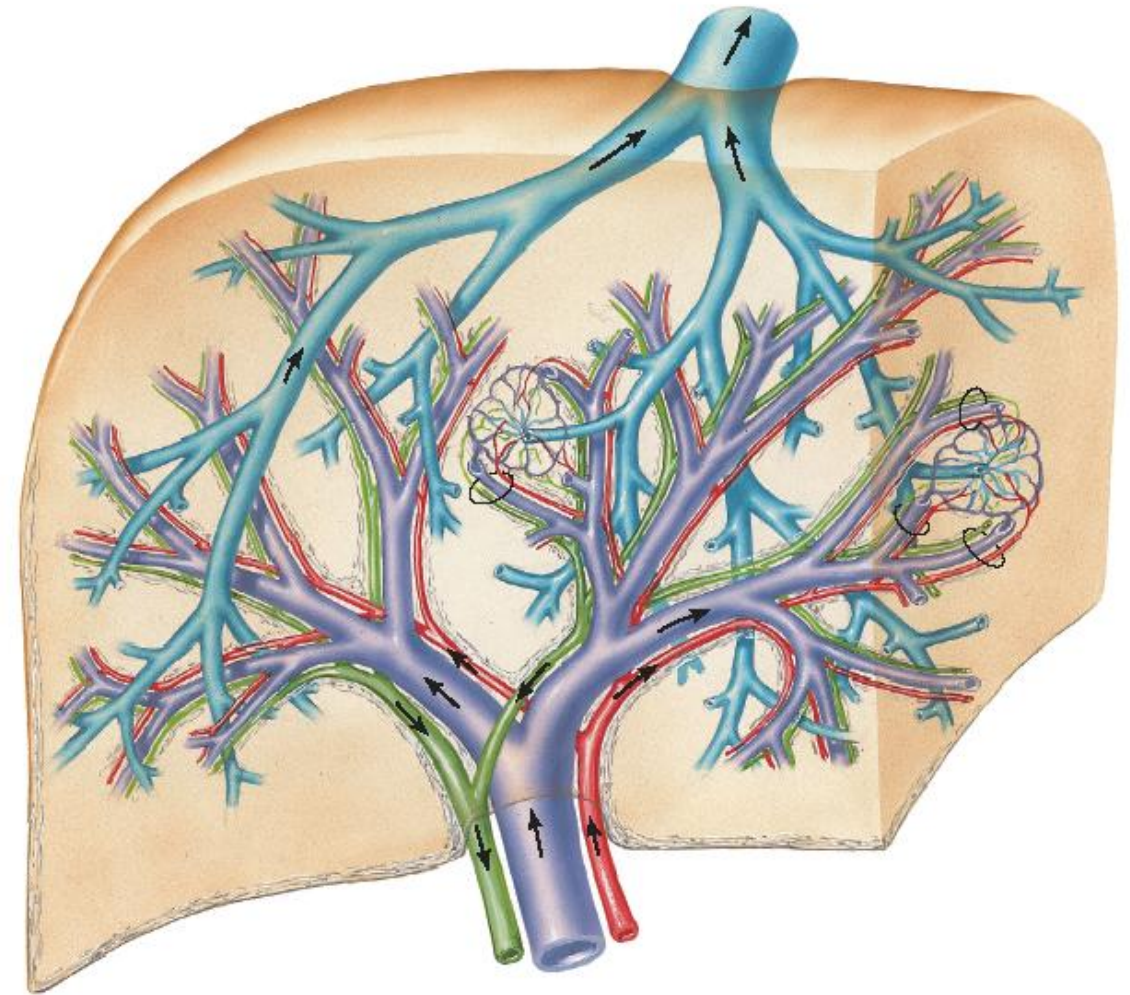
- Hepatic arteries, 30%

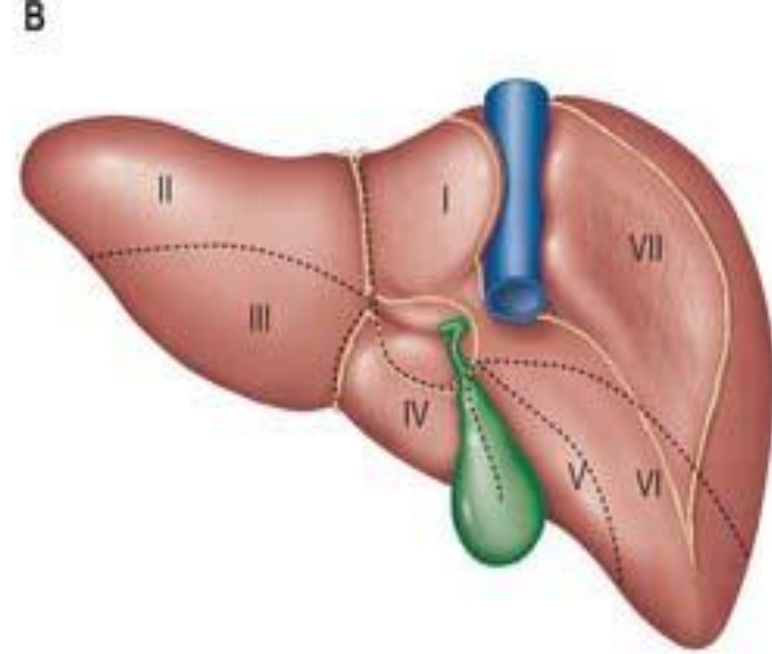
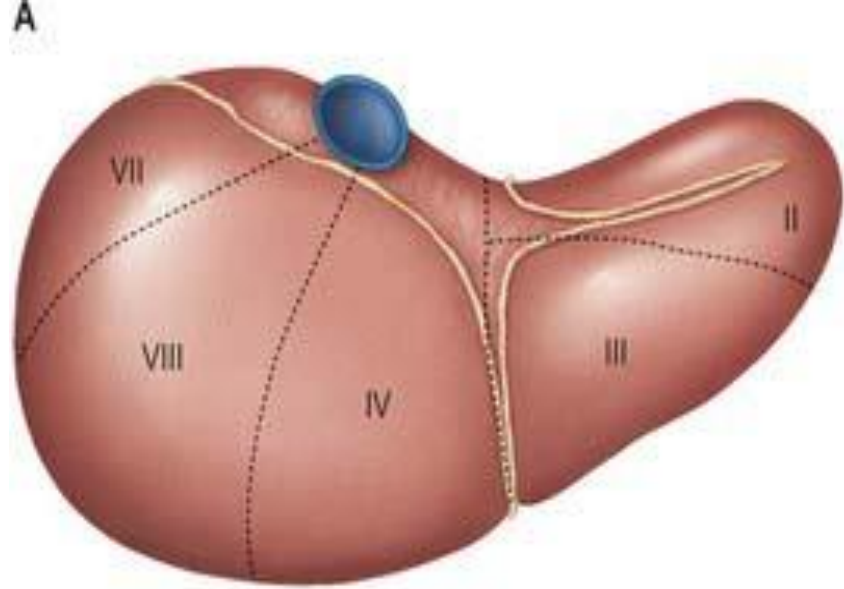
2. VENOUS :

- Portal vein, 70%
- Hepatic veins

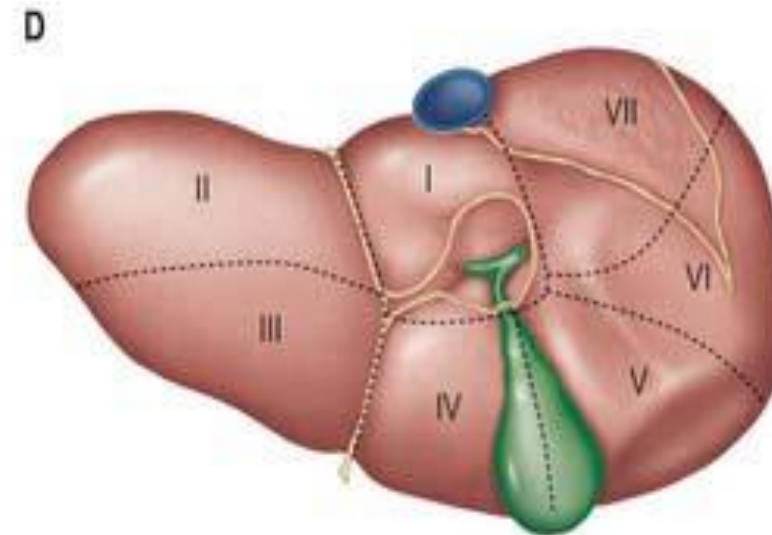
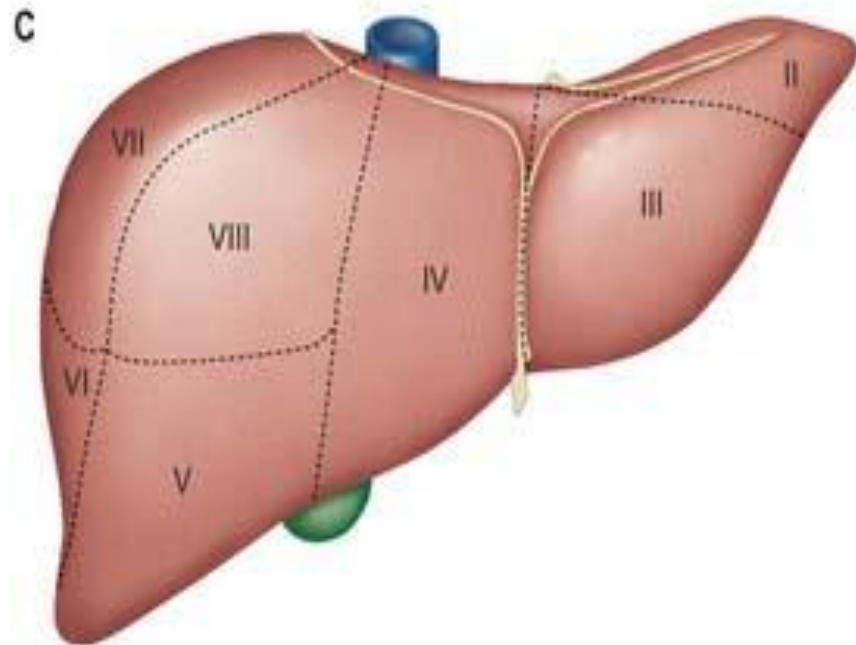
3. LYMPHAICS :

- $\frac{1}{2}$ of body lymph is formed by liver
- to LN in porta hepatis
→ celiac nodes
- from bare area through
diaphragm → post. Mediastinal LNs.

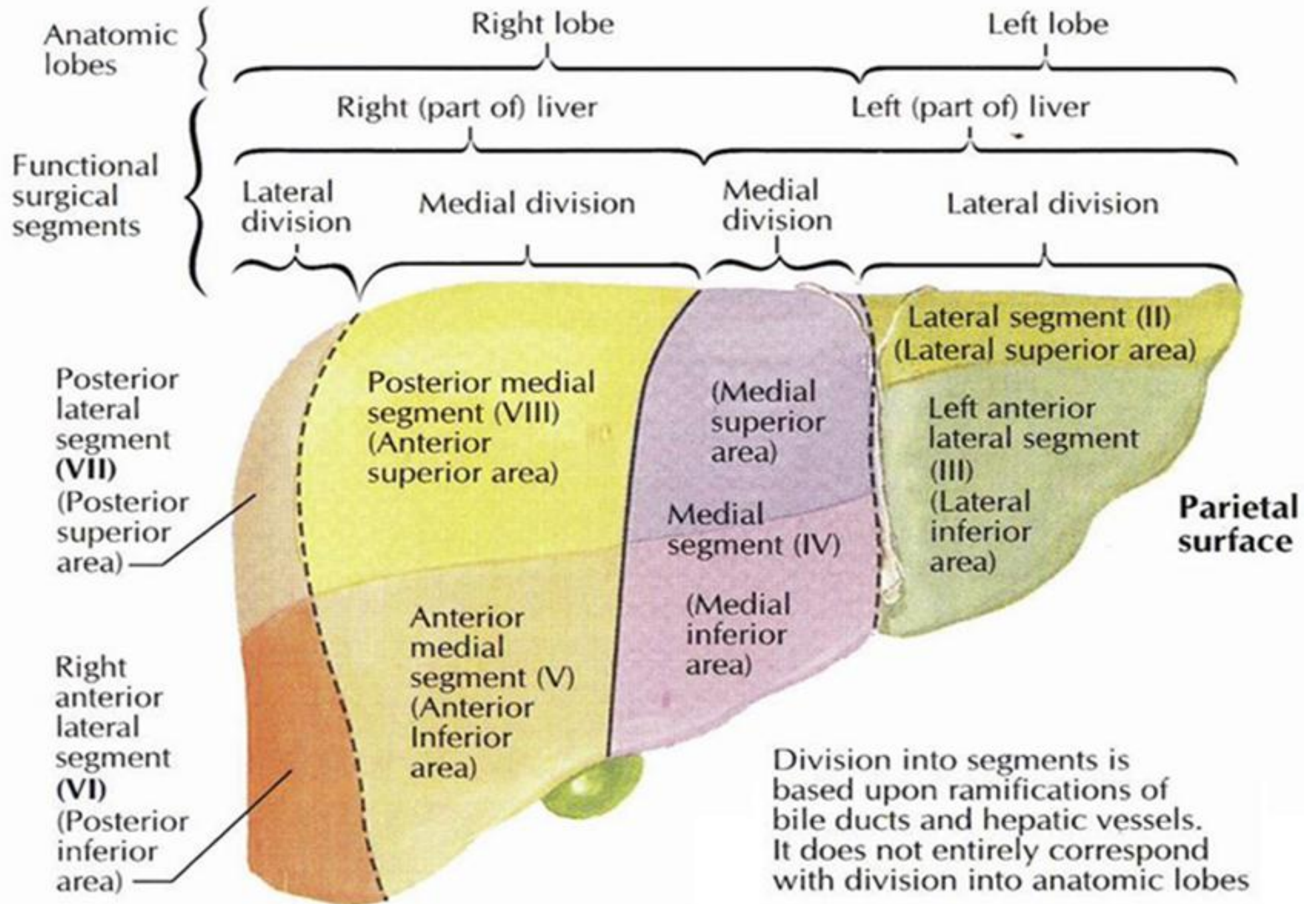




Hepatic segments



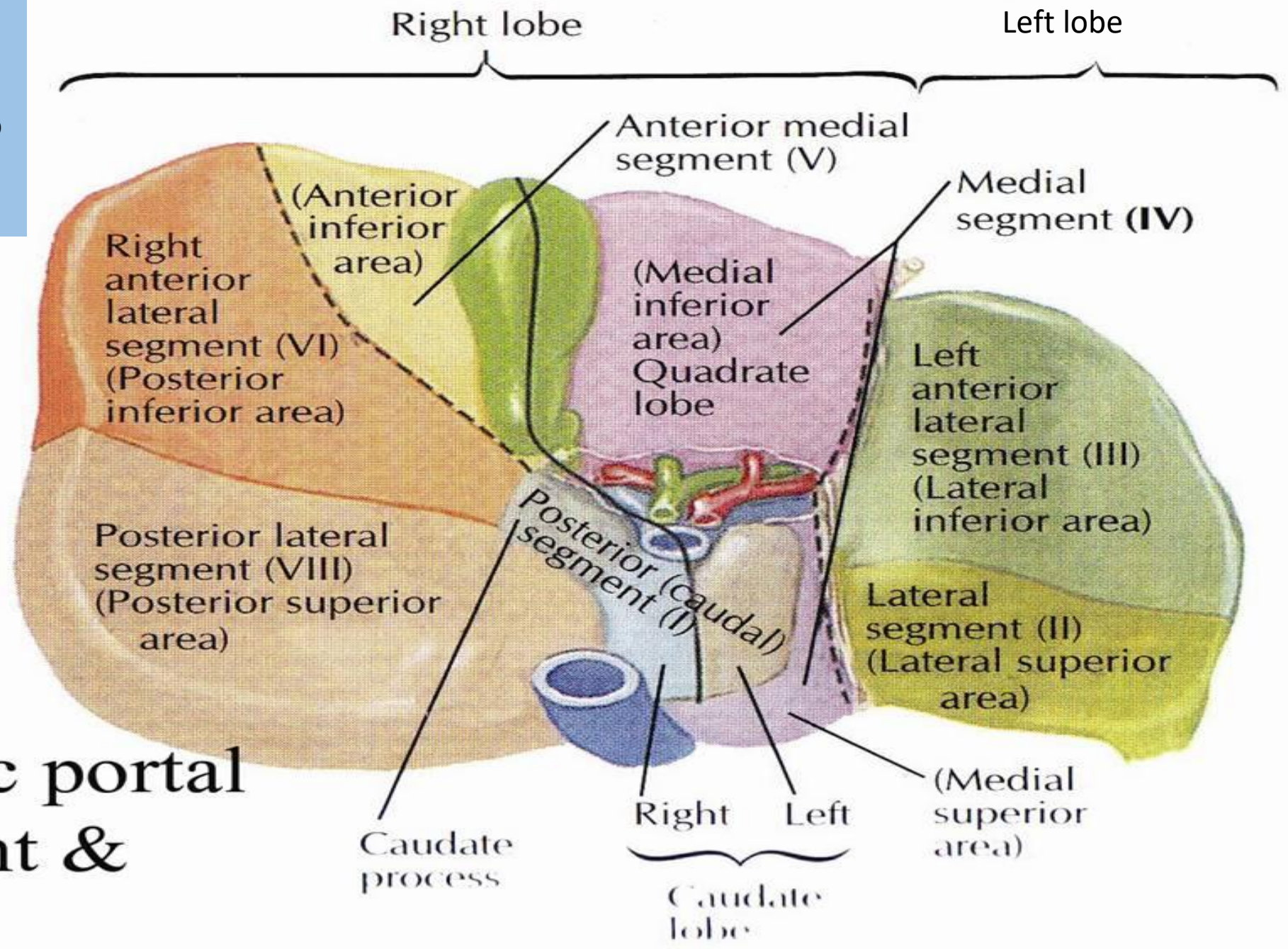
Hepatic Portal Segments & Lobules

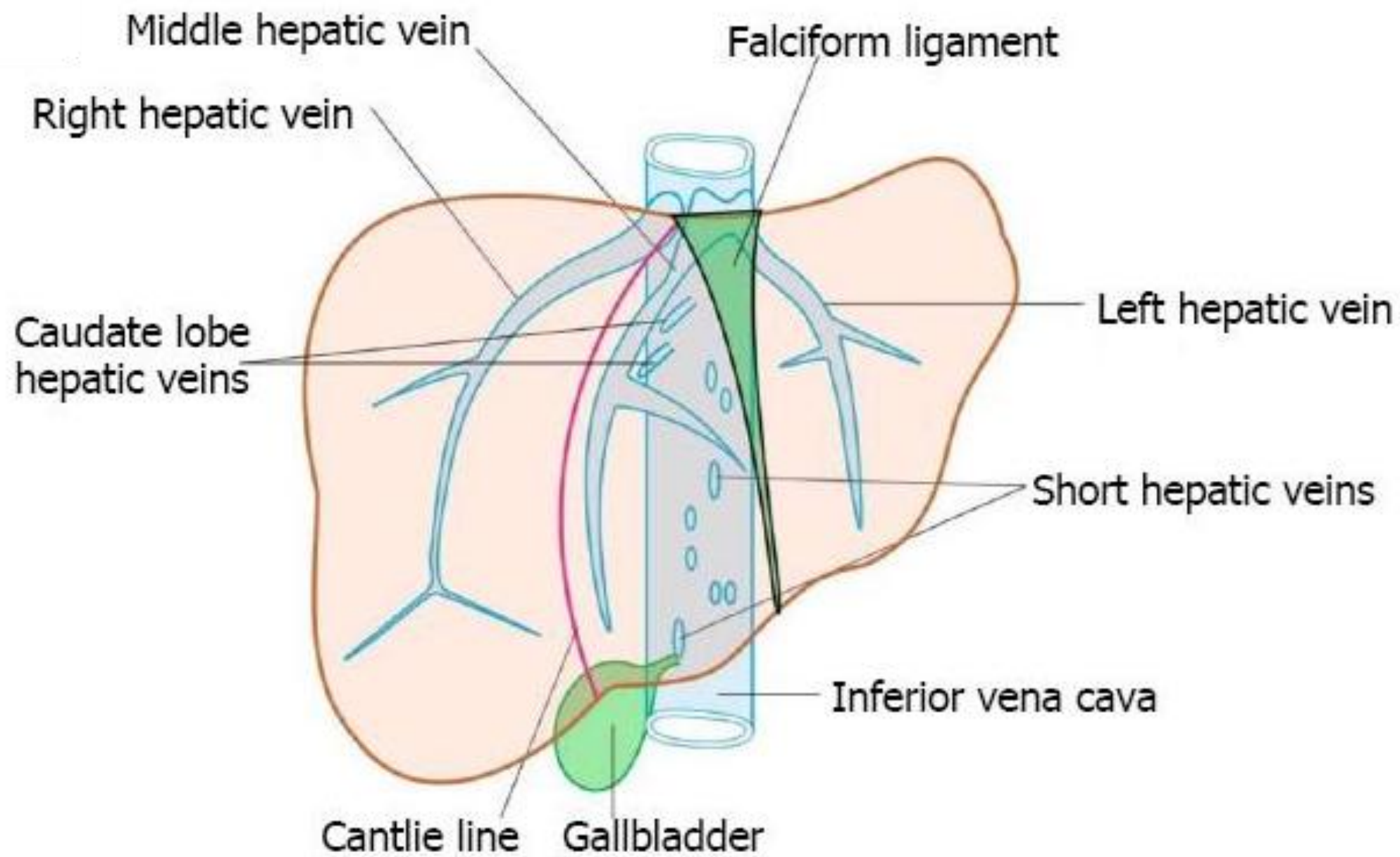


The line that divides the liver into 2 functional (physiological) lobes is called **Cantlie's line** (also known as the **principal plane** of the liver).

Visceral surface

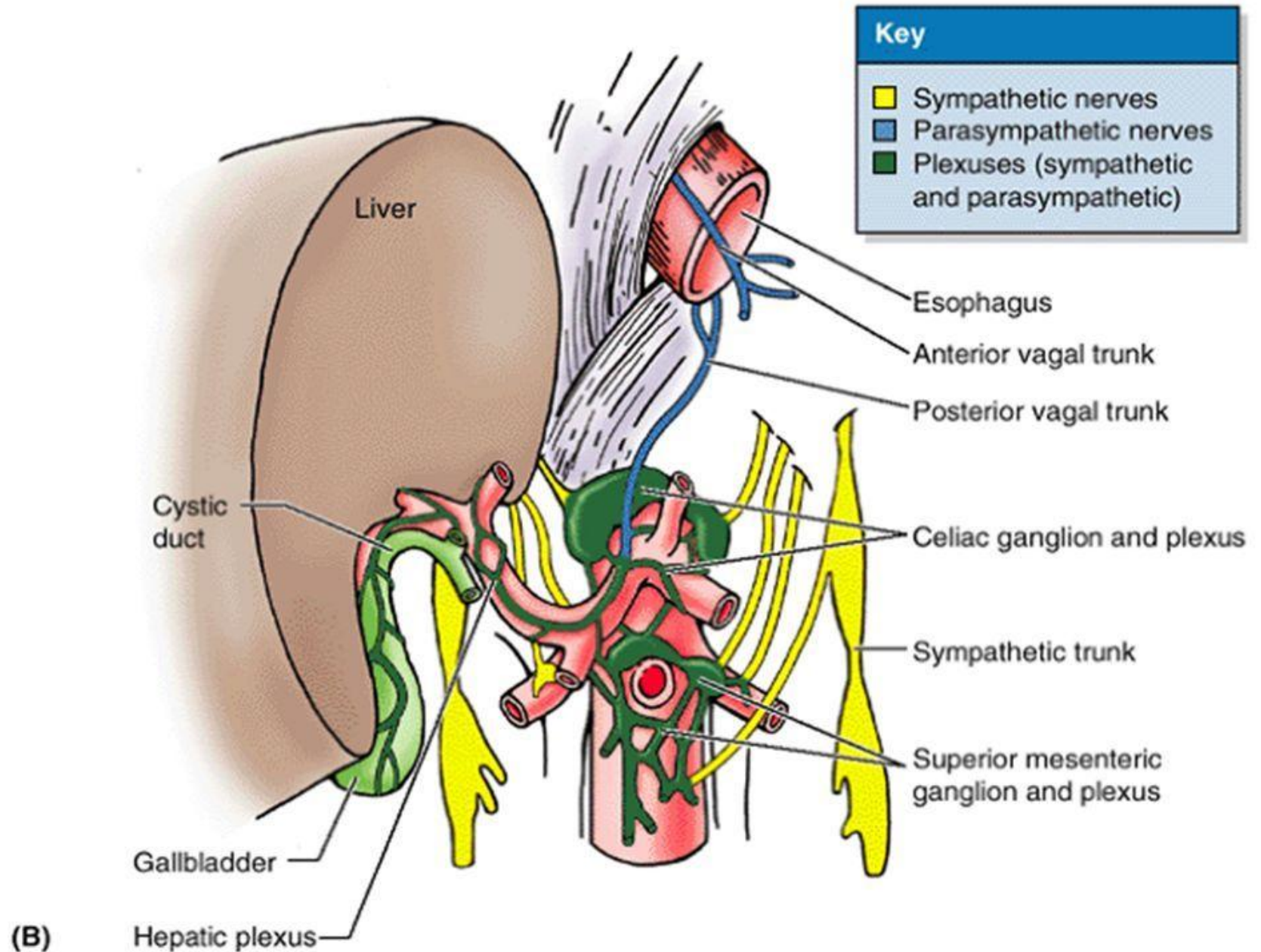
Hepatic portal segment & lobule





4. NERVE SUPPLY :

- Symp. & Parasymp. from coeliac plexus
- Anterior vagal trunk gives a hepatic branch to liver.



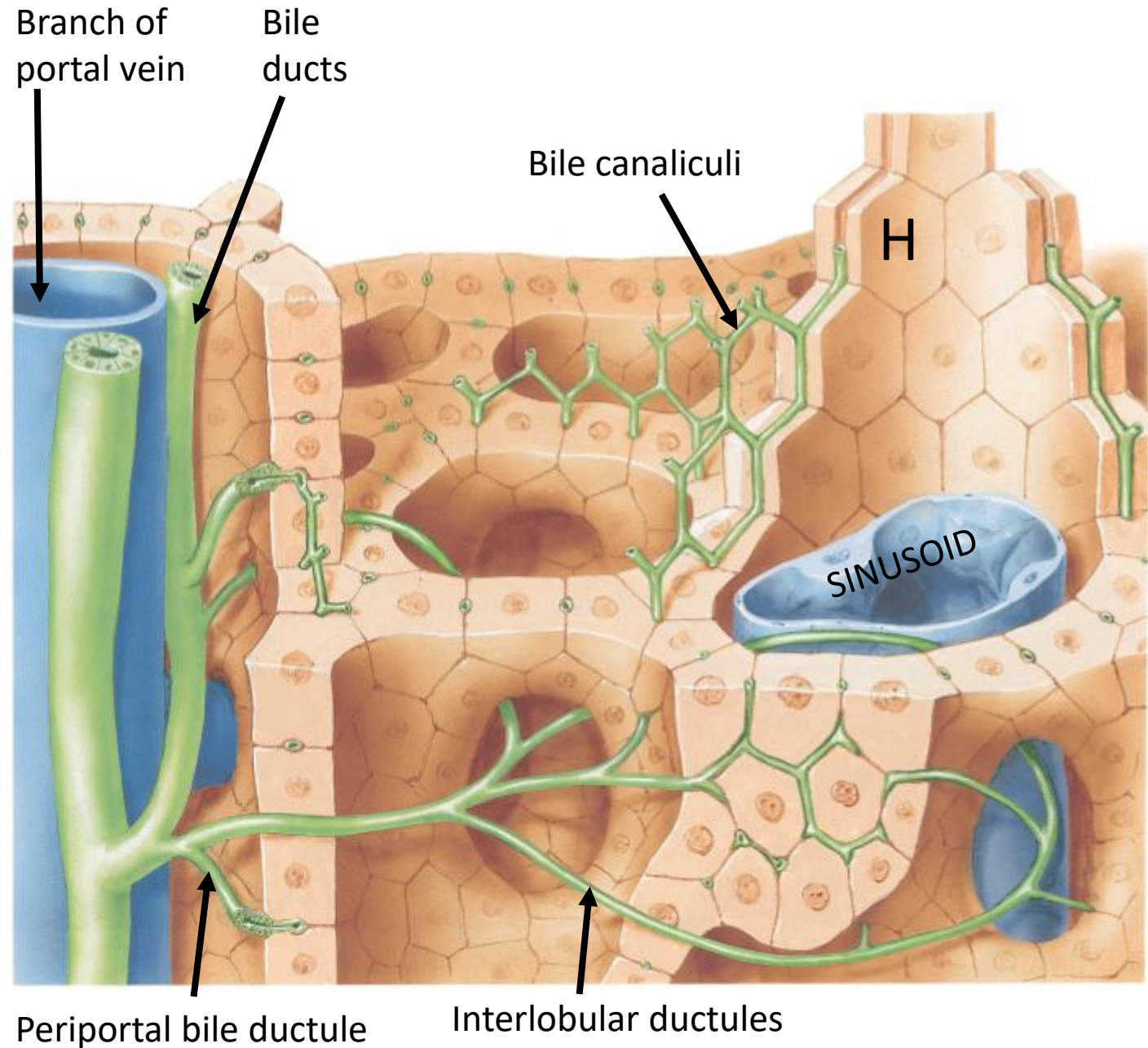
Clinical points:

- Liver support
- Liver trauma
- Liver biopsy
- Sub-phrenic spaces

BILIARY SYSTEM:

- INTRAHEPATIC BILIARY SYSTEM

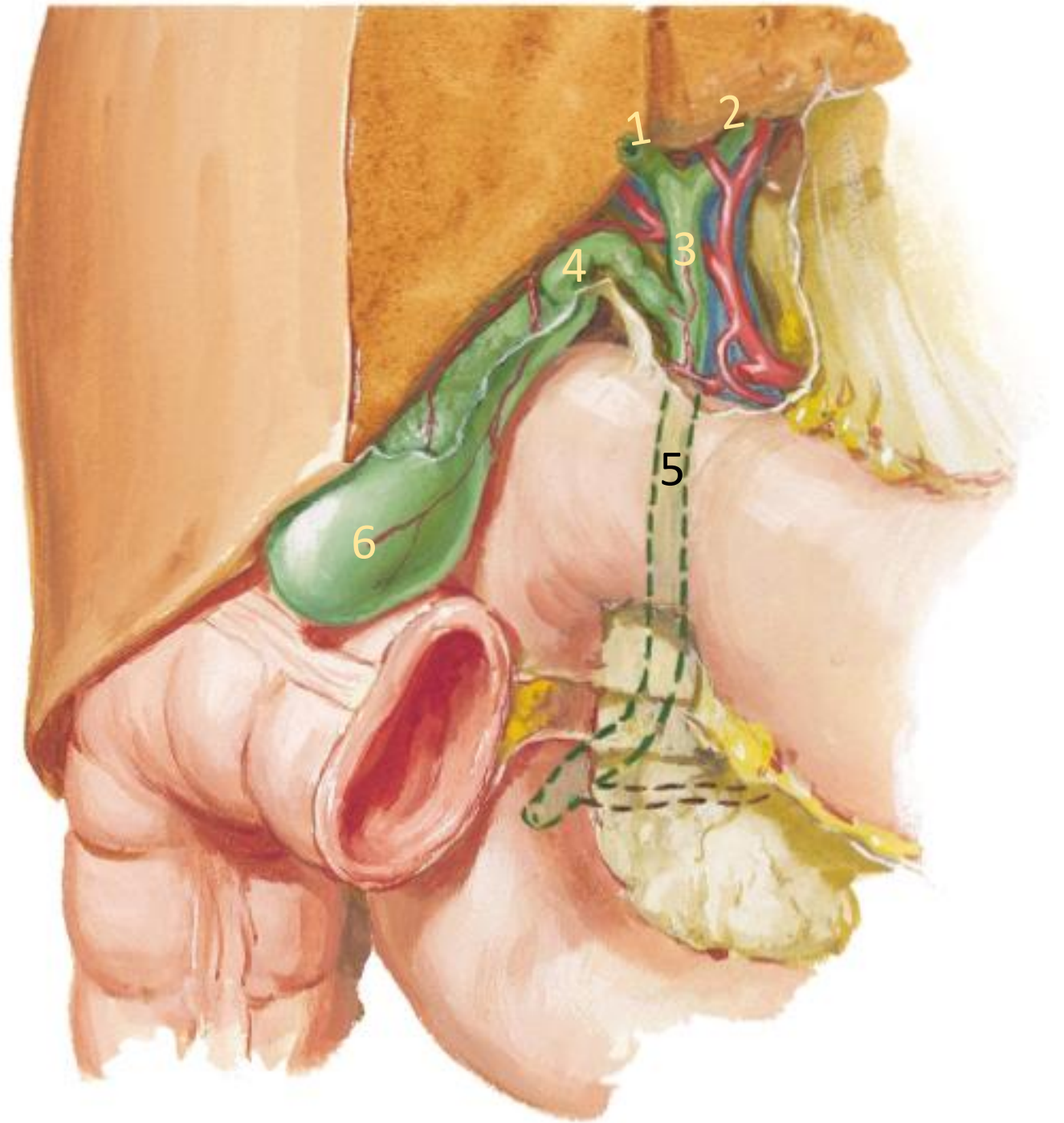
NOT
OUR BUISNESS
!!!!



- EXTRAHEPATIC
BILIARY SYSTEM

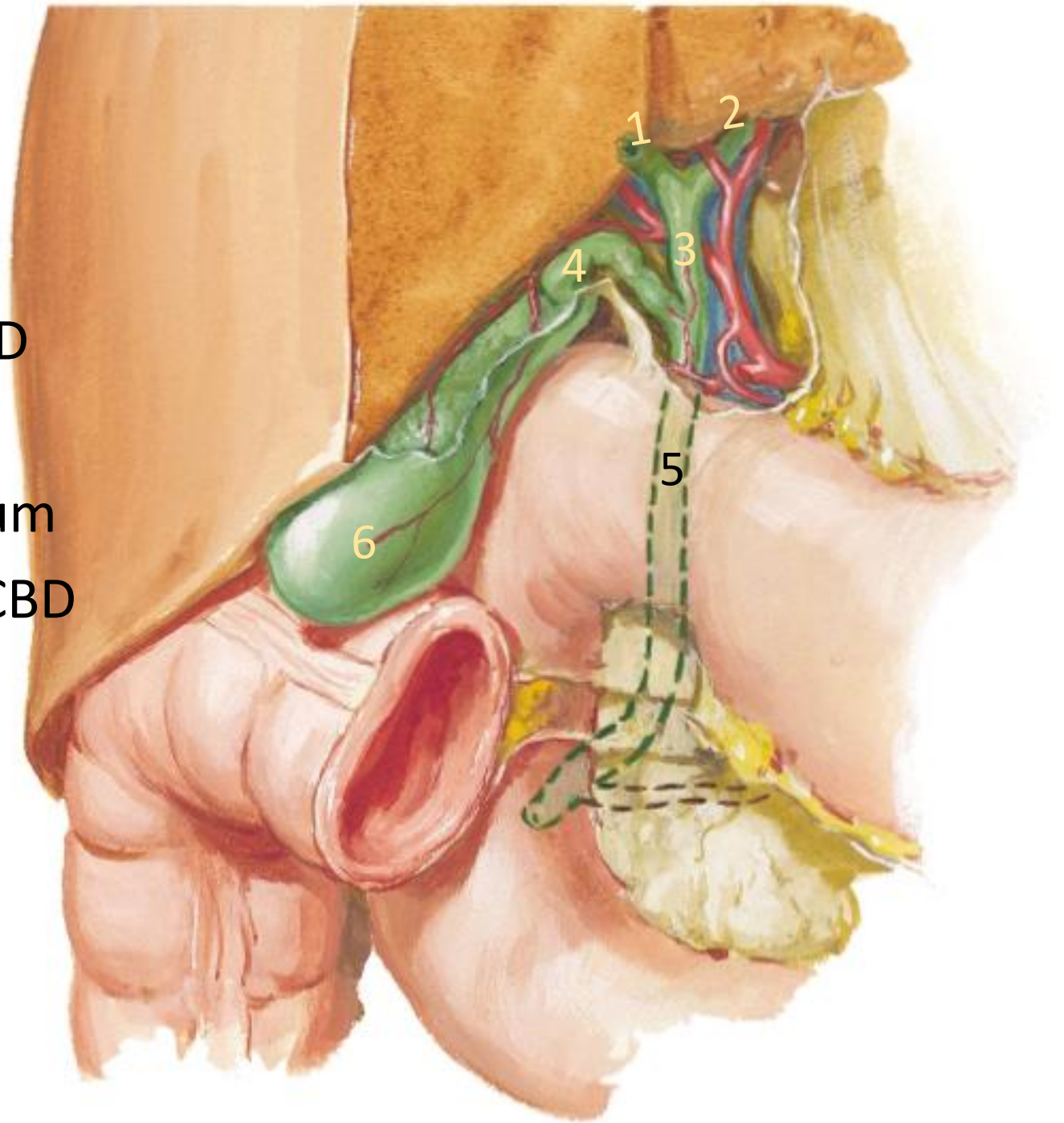
Consists of :

1. Rt. hepatic duct
2. Lt. hepatic duct
3. Com. hepatic duct
4. Cystic duct
5. Com. Bile duct
6. Gallbladder



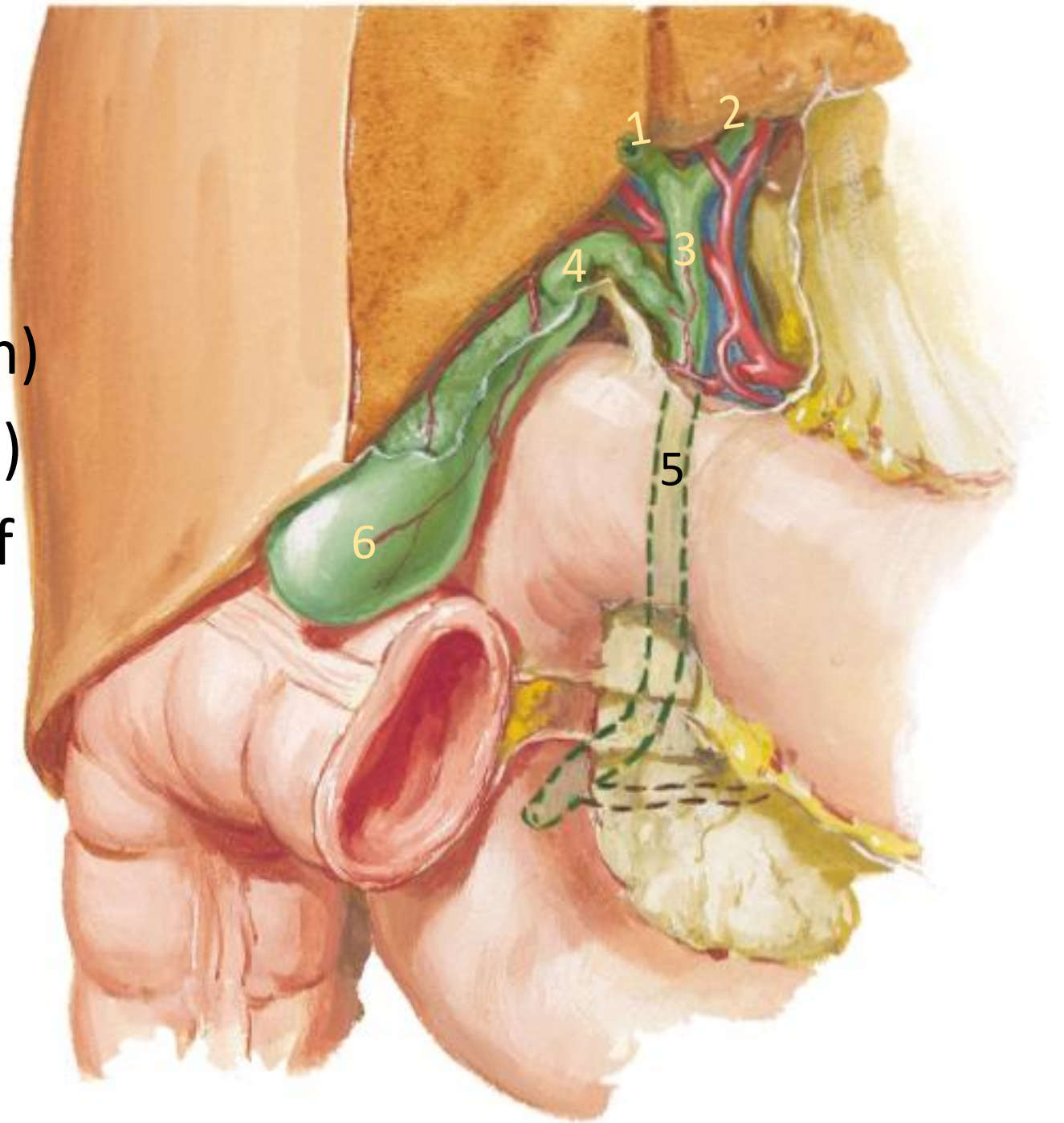
HEPATIC DUCTS :

- Emerge from porta hepatis
- Rt. & Lt. HDs unite to form CHD
- CHD = 4 cm
- CHD descend in lesser omentum
- Joined by cystic duct to form CBD

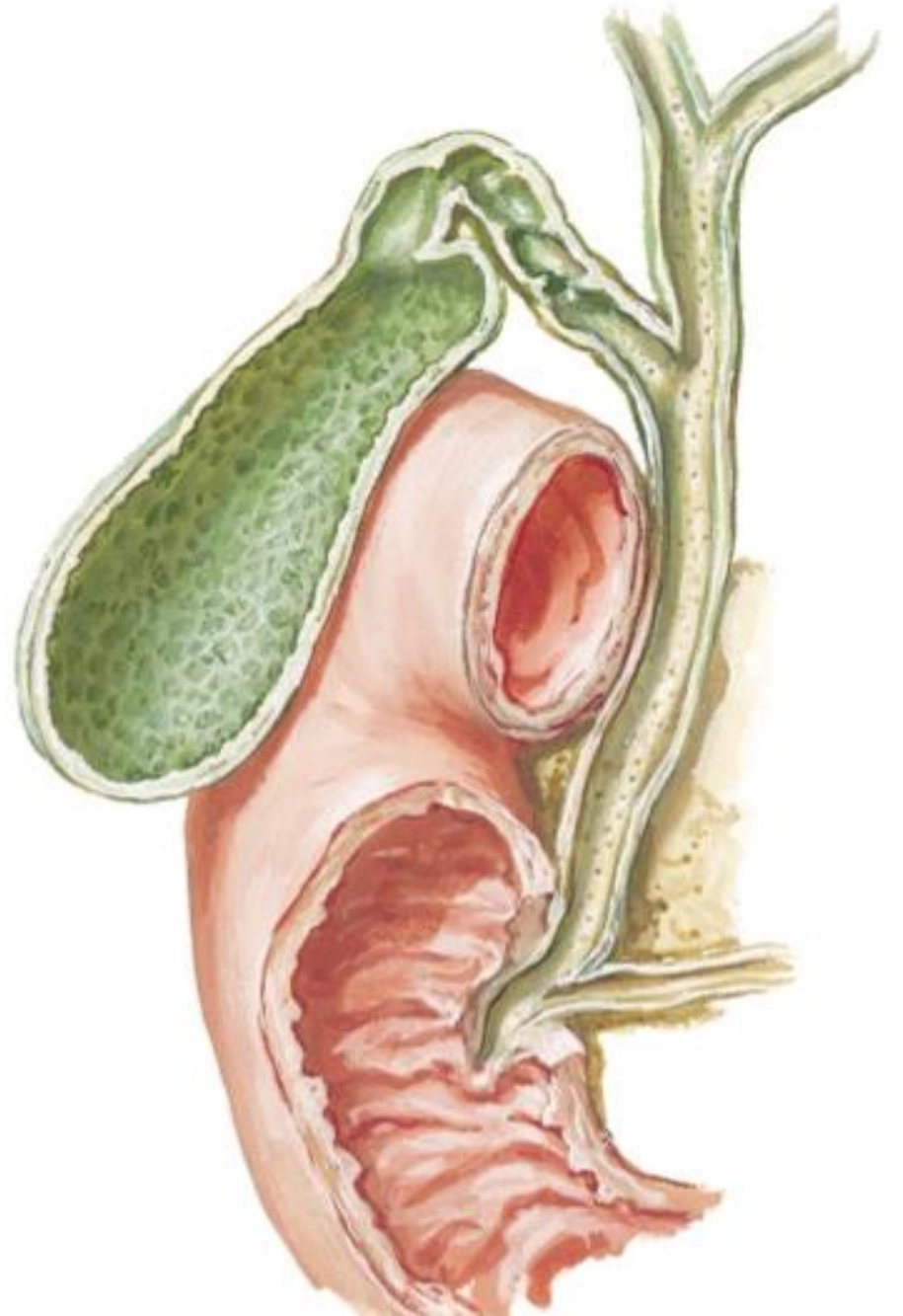
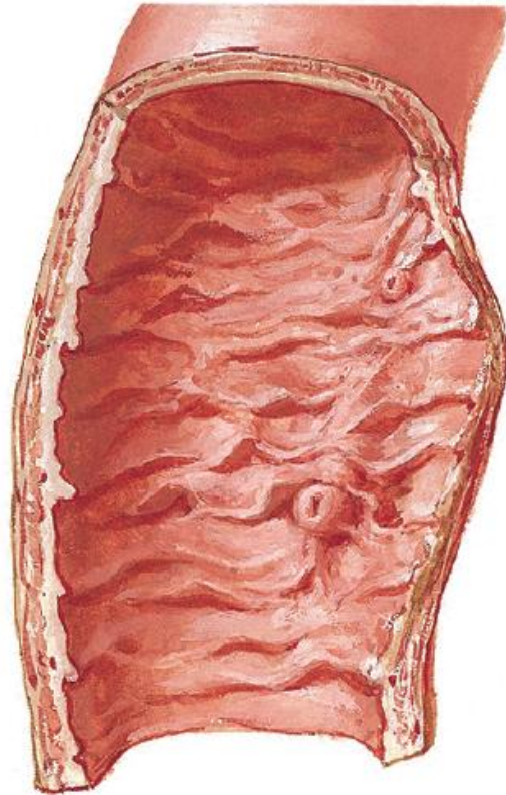
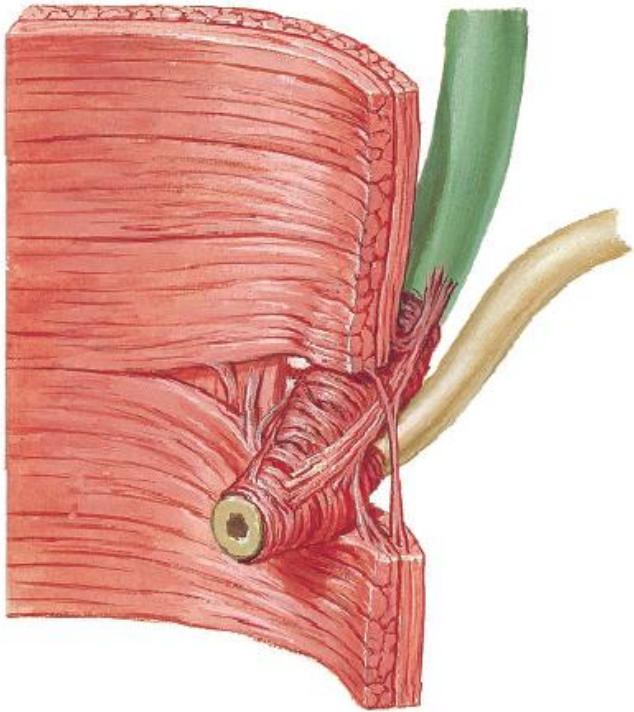


COMMON BILIARY DUCT:

- 8cm long
- 1st course (Lesser Omentum)
- 2nd course (behind 2nd Duo.)
- 3rd course (post. To head of pancreas)

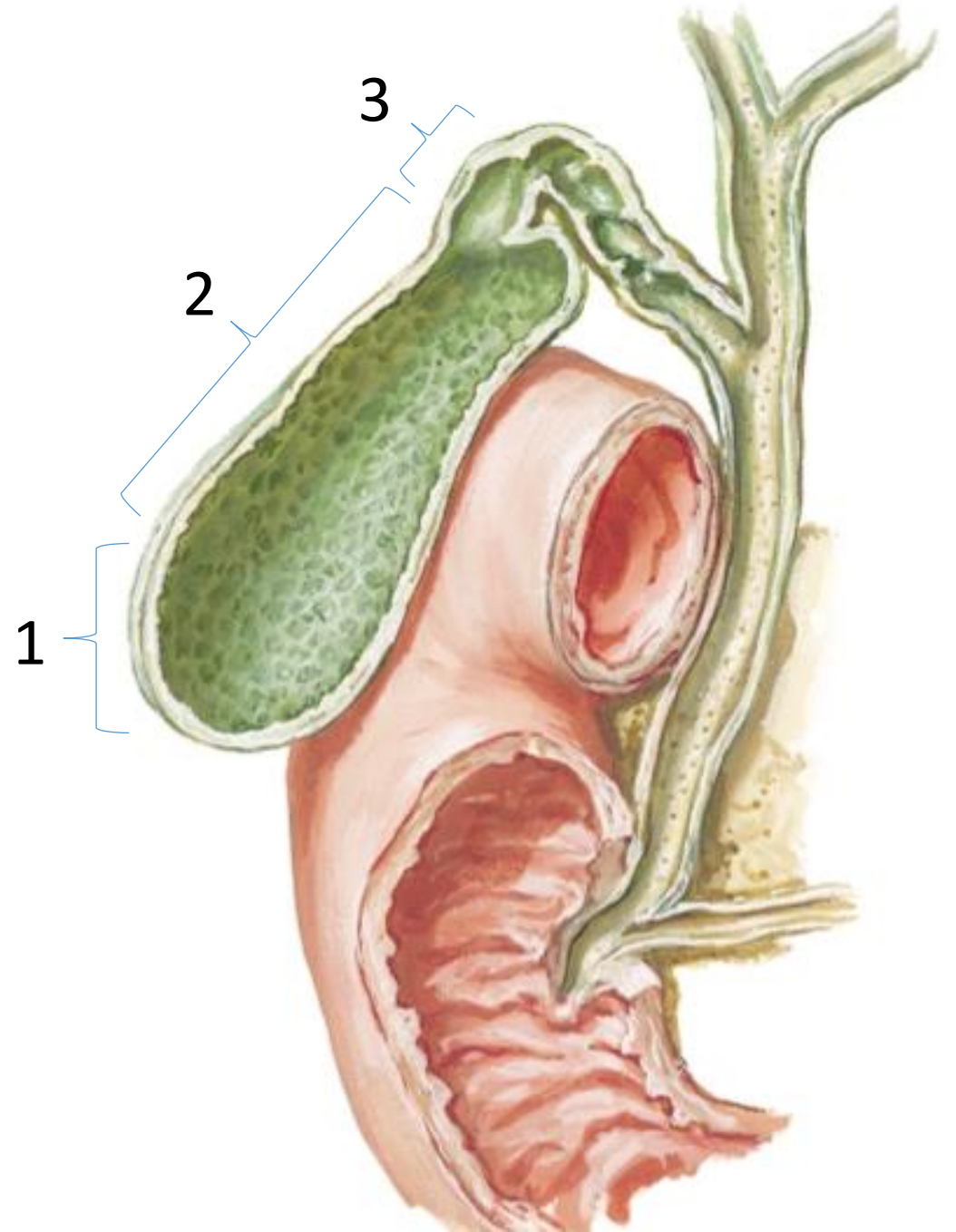


- Hepatopancreatic ampulla of Vater
- Major duodenal papilla
- Sphincter of Oddi



GALLBLADDER:

- Under the liver
- Stores 30-50 ml of bile
- Function
- Divided into :
 1. Fundus
 2. Body
 3. Neck



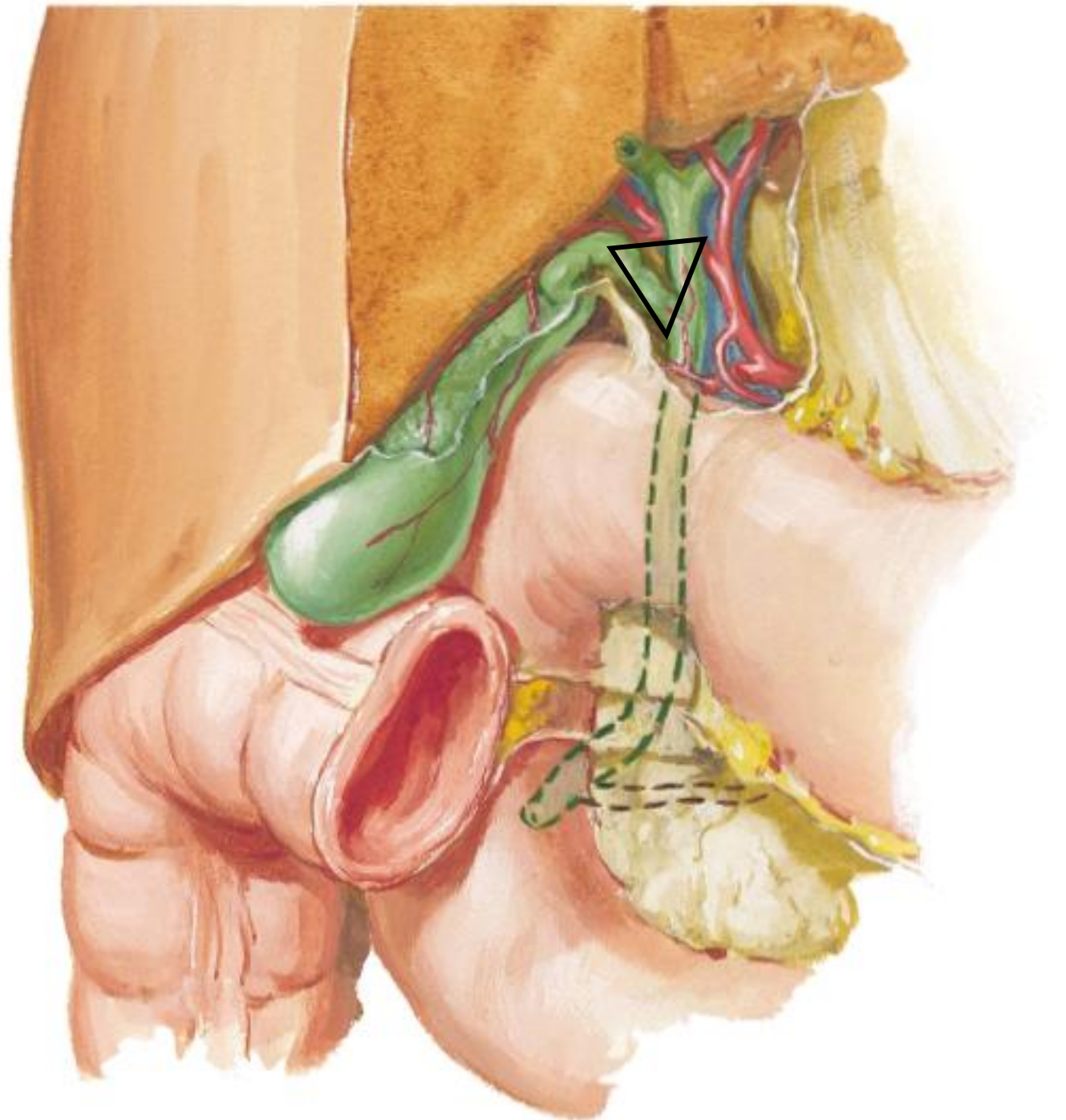
RELATIONS OF GALLBLADDER:

ANTERIORLY:

- Ant. Abdominal wall
- Inferior surface of liver

POSTERIORLY:

- Transverse colon
- 1st & 2nd duodenum



BLOOD SUPPLY:

ARTERIAL :

- Cystic artery

VENOUS :

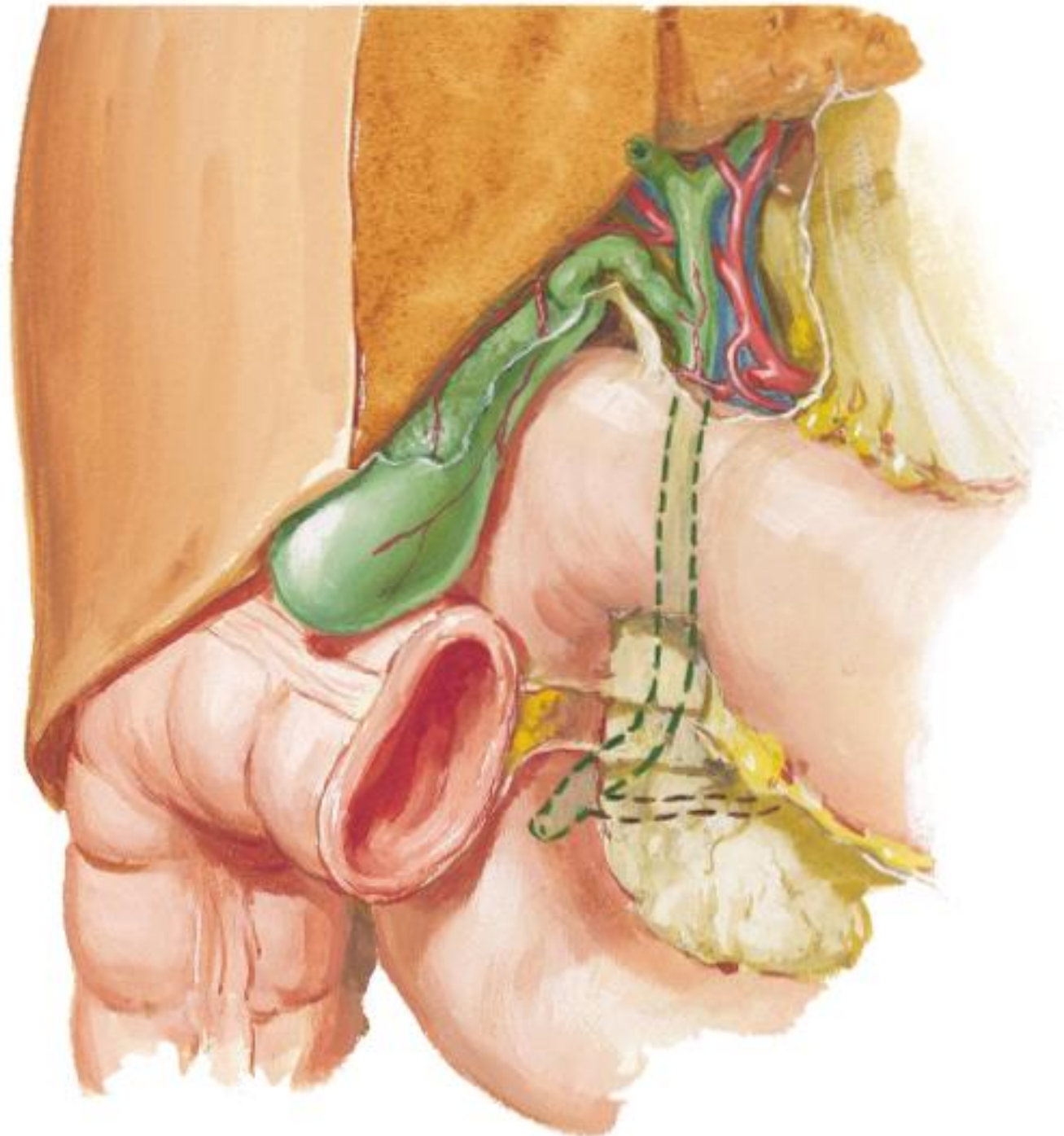
- Cystic vein

LYMPHATICS :

- A cystic LN at the neck of GB

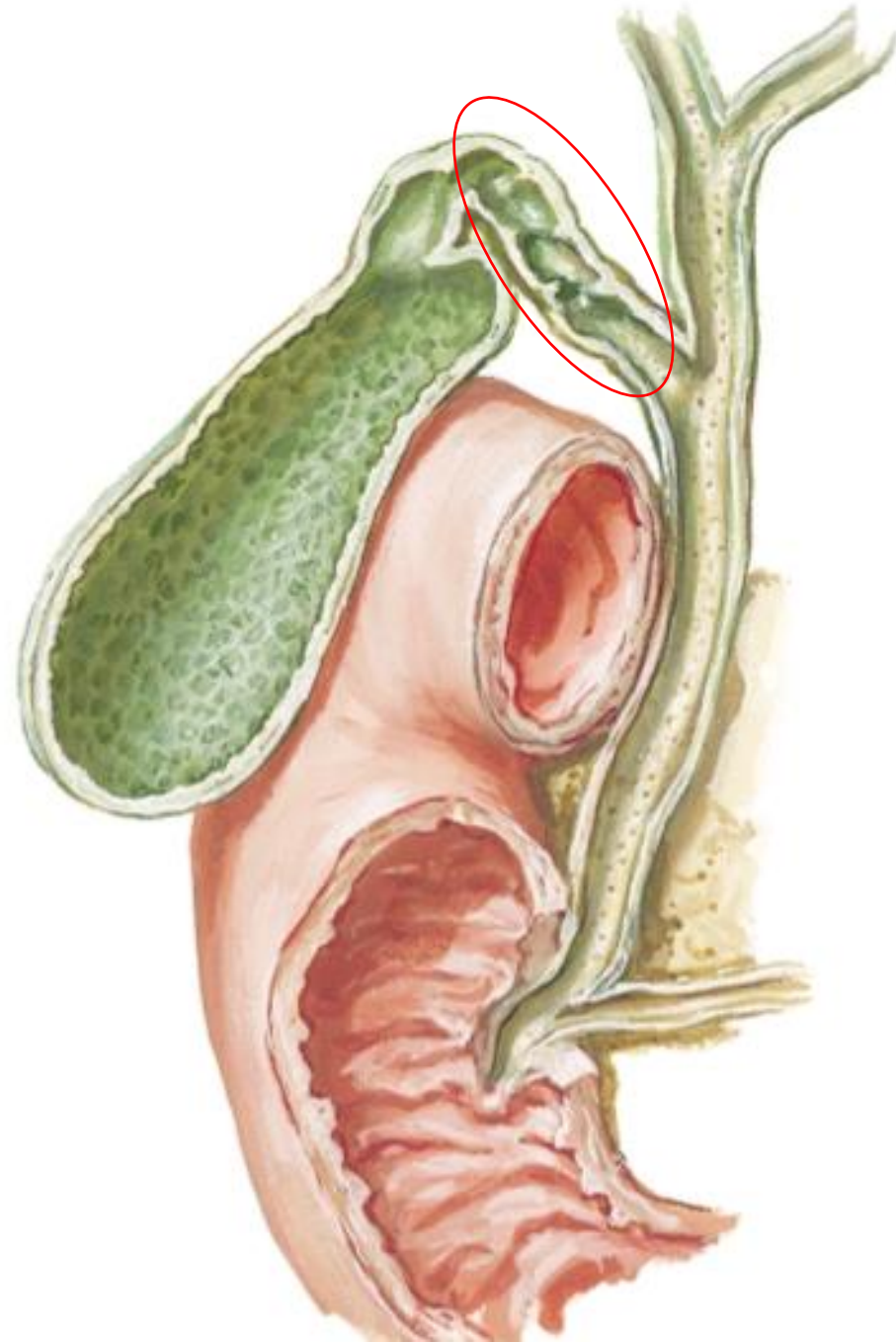
NERVES :

- Autonomic from coeliac plexus
- Cholecystokinin

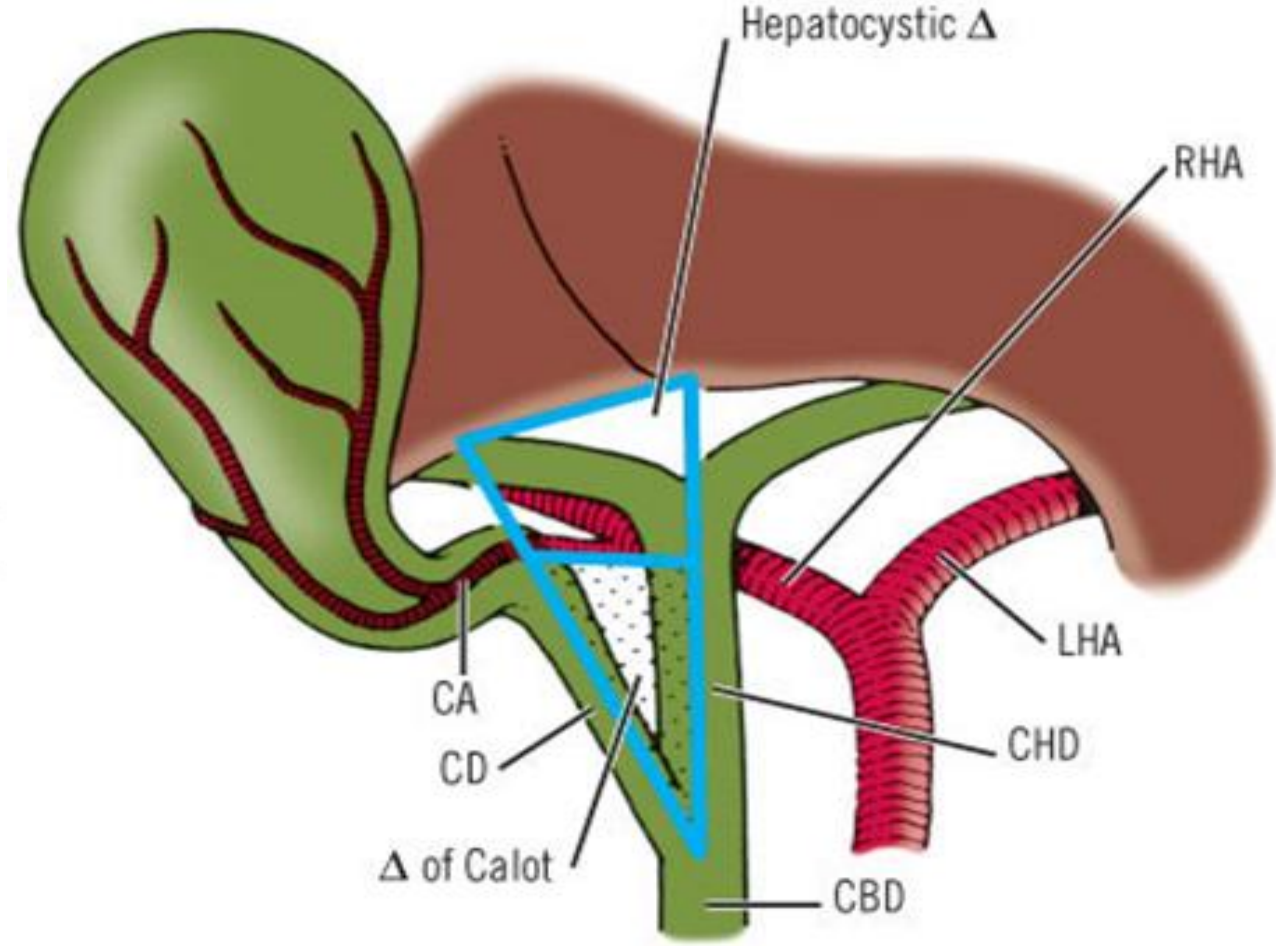


CYSTIC DUCT

- 3.8 cm
- Somewhat S shaped
- Its mucous membrane is forming a spiral fold (the spiral valve)



- **Calot's triangle**
bounded by cystic duct, cystic artery, and common hepatic duct.
- **Hepatocystic triangle**
bounded by gallbladder wall and cystic duct, liver edge, and common hepatic duct; the cystic artery (and hence Calot's triangle) lies within this space.

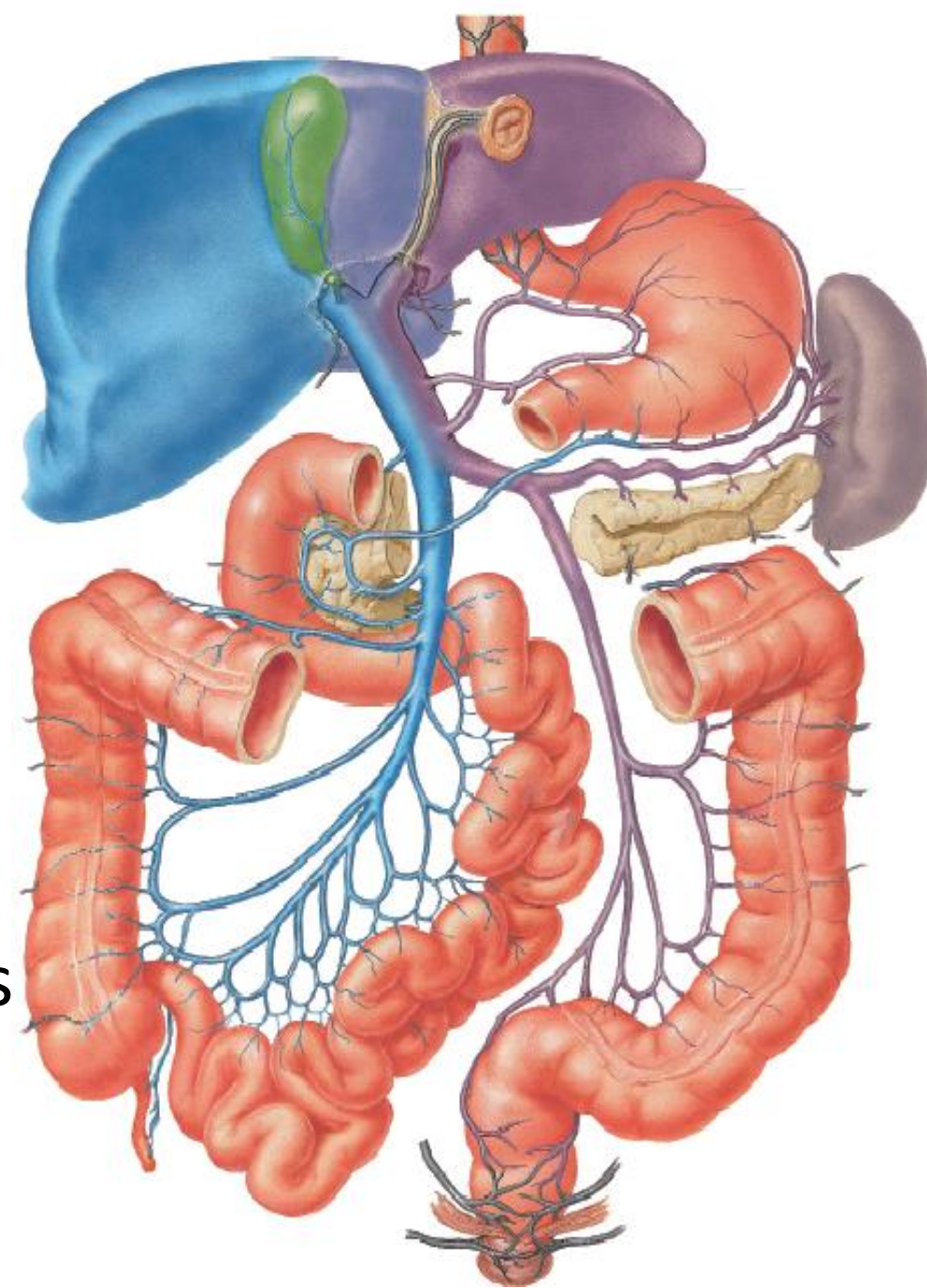


Clinical points:

- Gallstones
- Biliary colic (epigastric pain, jaundice & pancreatitis)
- Acute cholecystitis & referred pain to shoulder
- Cholecystectomy (variations)
- Gangrenous (single arterial supply)
- Biliary atresia

PORTAL VEIN:

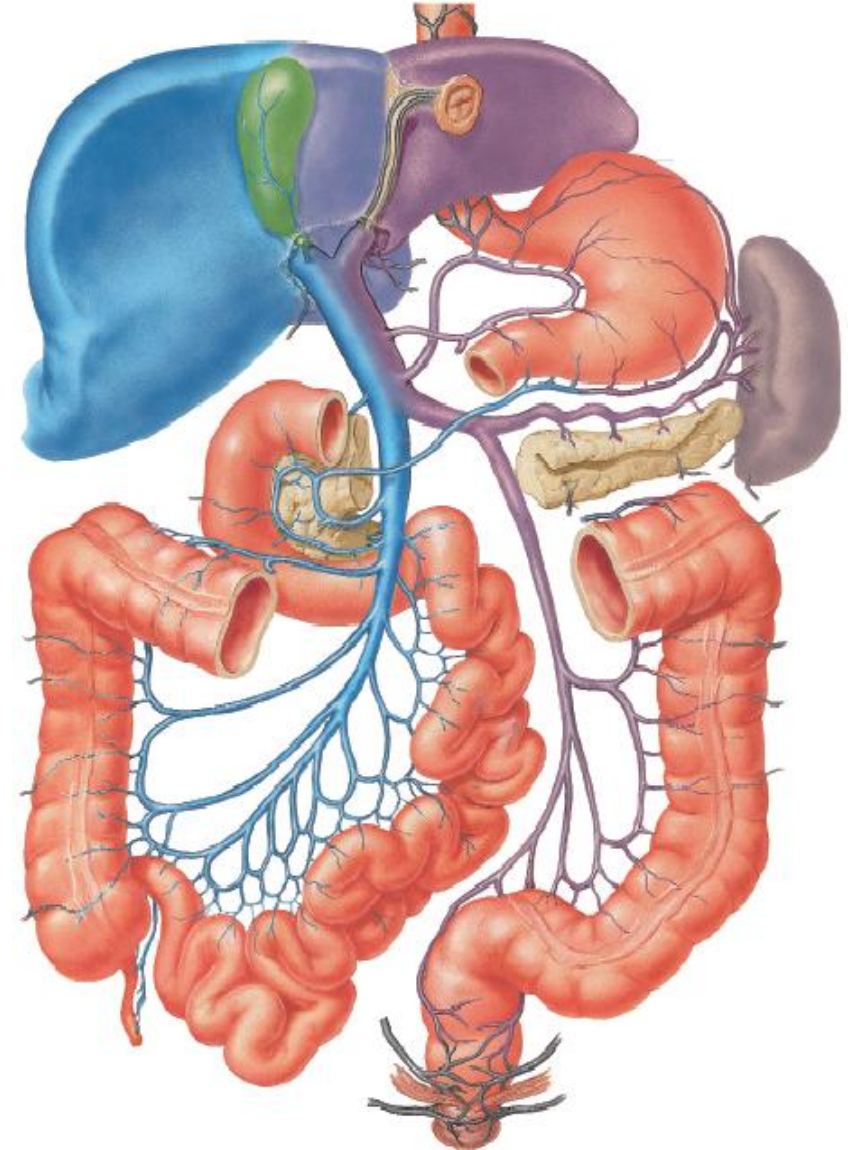
- about 2 in. (5 cm) long
- Drains blood from:
 1. abdominal part of the GIT
 2. lower 1/3 of esophagus
 3. upper 1/2 of anal canal
 4. spleen, pancreas and gallbladder
- Formed behind the neck of the pancreas by the union of SMV & SPV



- Ascends to the right, behind the first part of the duodenum, and enters lesser omentum.
- Runs upward in front of the opening into the lesser sac to the porta hepatis where it divides in to:

Right and Left Terminal Branches.

- Tributaries :
 1. Splenic vein,
 2. superior mesenteric vein,
 3. left gastric vein,
 4. right gastric vein,
 5. cystic veins



PORTO-SYSTEMIC ANASTOMOSIS :

1. Lower esophagus:

portal : Lt. gastric vein

caval : azygos vein

2. Upper anal canal :

portal : sup. Rectal vein

caval : middle & inferior rectal veins

3. Umbilical :

portal : paraumbilical vein

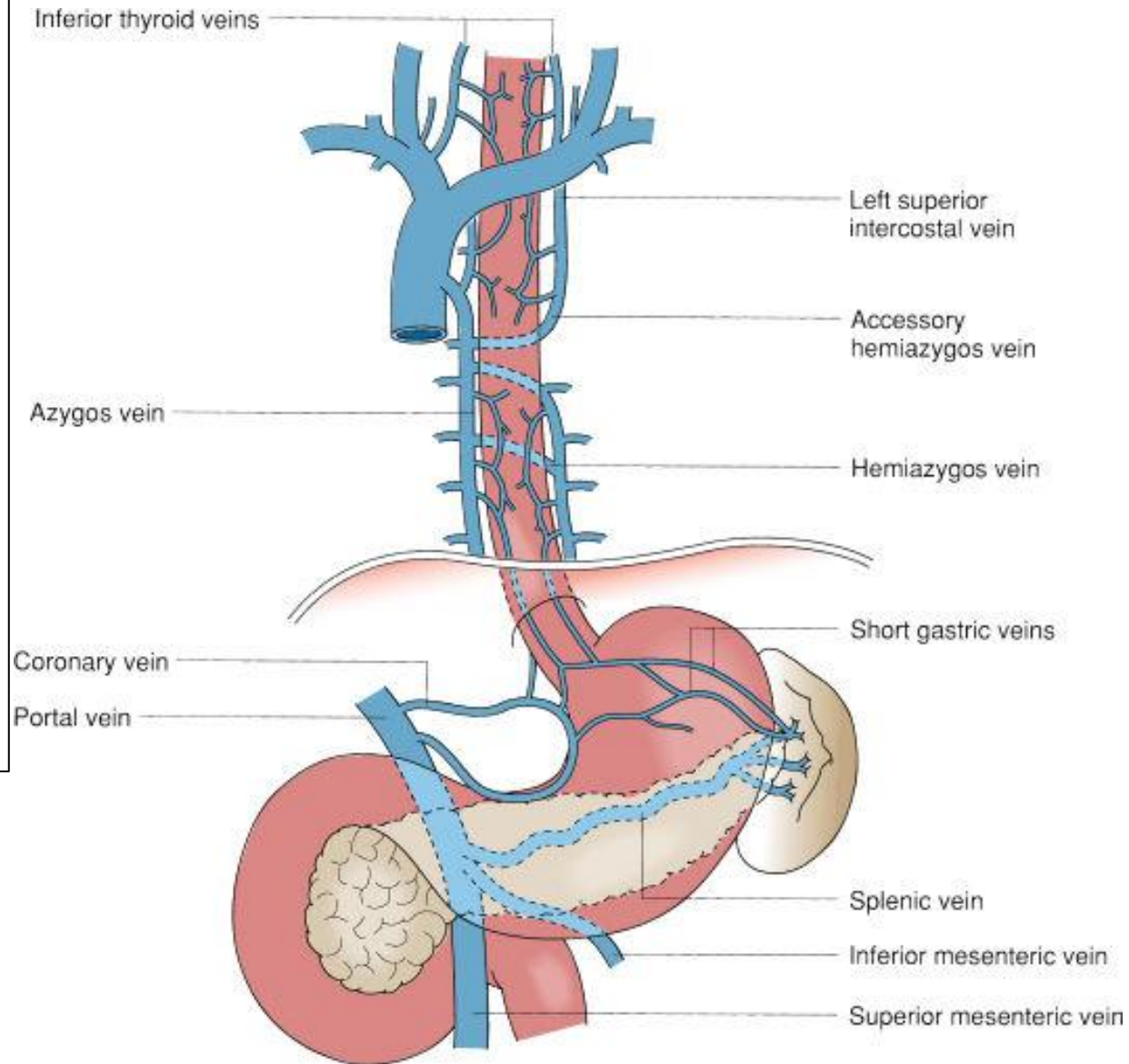
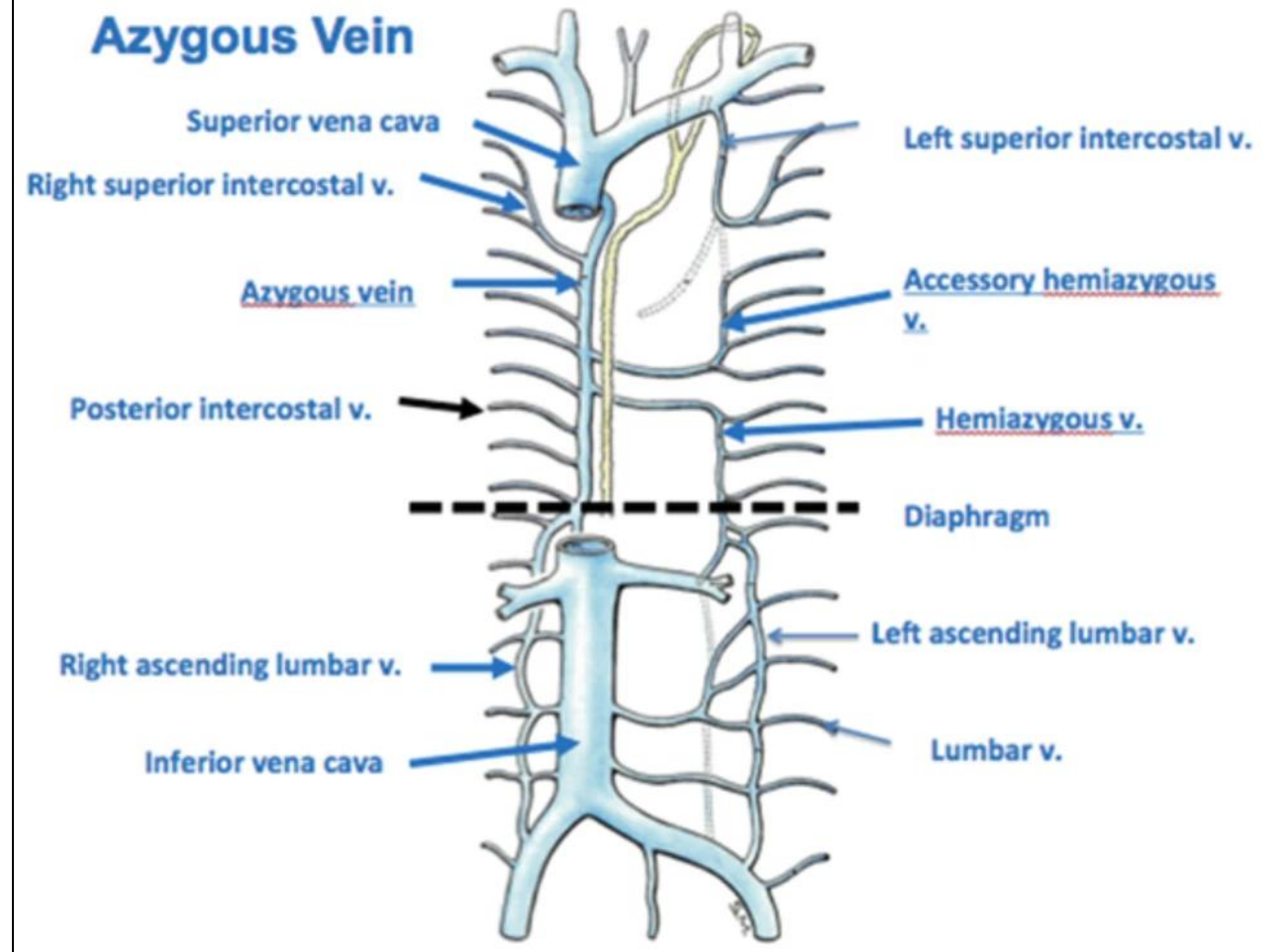
caval : superficial veins of the ant. abd. wall.

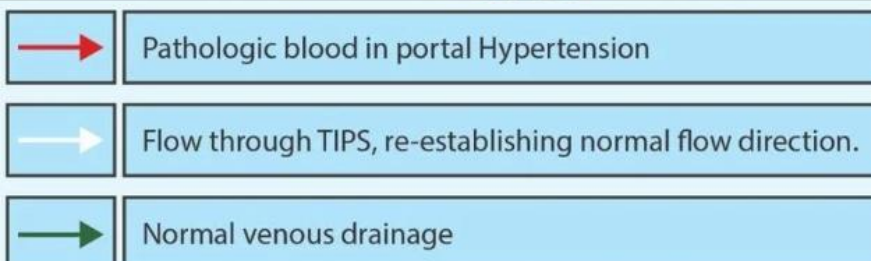
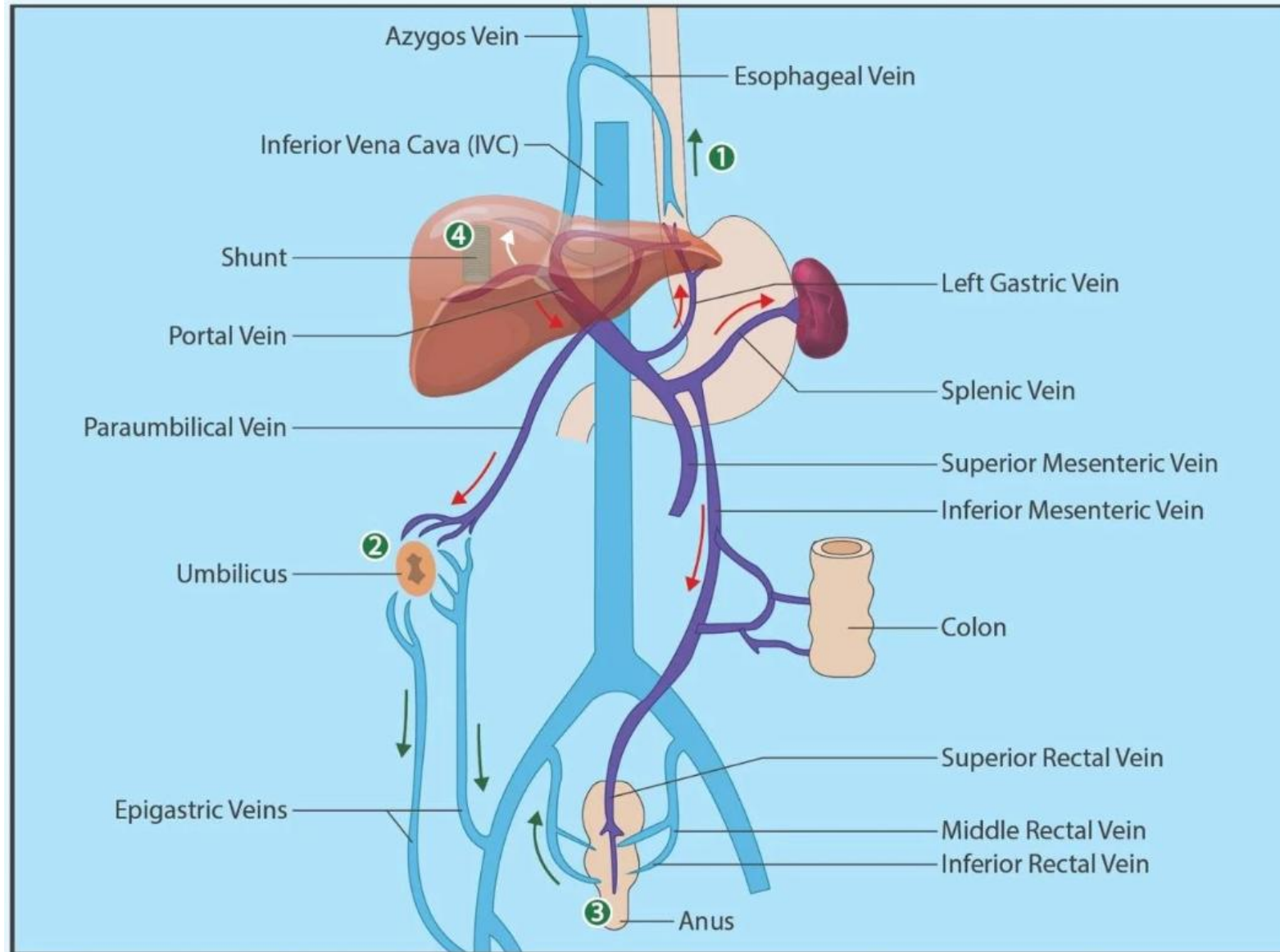
4. Retroperitoneal :

portal : colonic, duodenal, pancreatic , bare area

caval : lumbar, renal veins & inf. phrenic veins

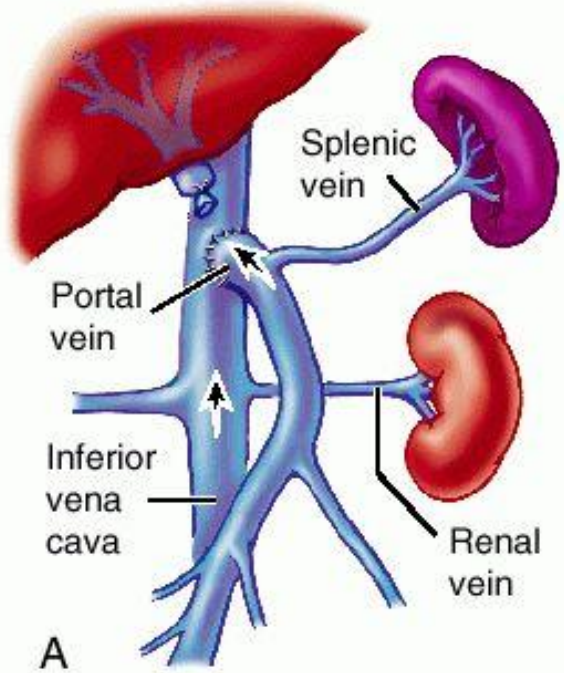
Azygous Vein





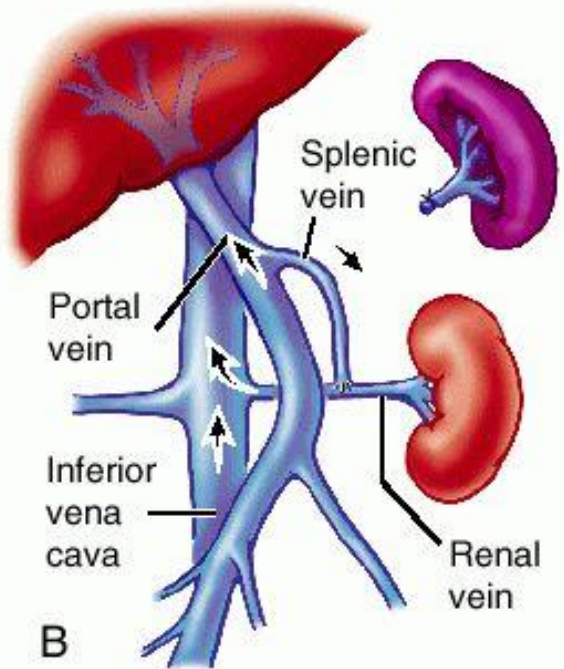
 Systemic Venous System

 Portal Venous System



Clinical points

- Portal hypertension
- Portocaval shunt
- Portal vein & multiple secondaries in liver



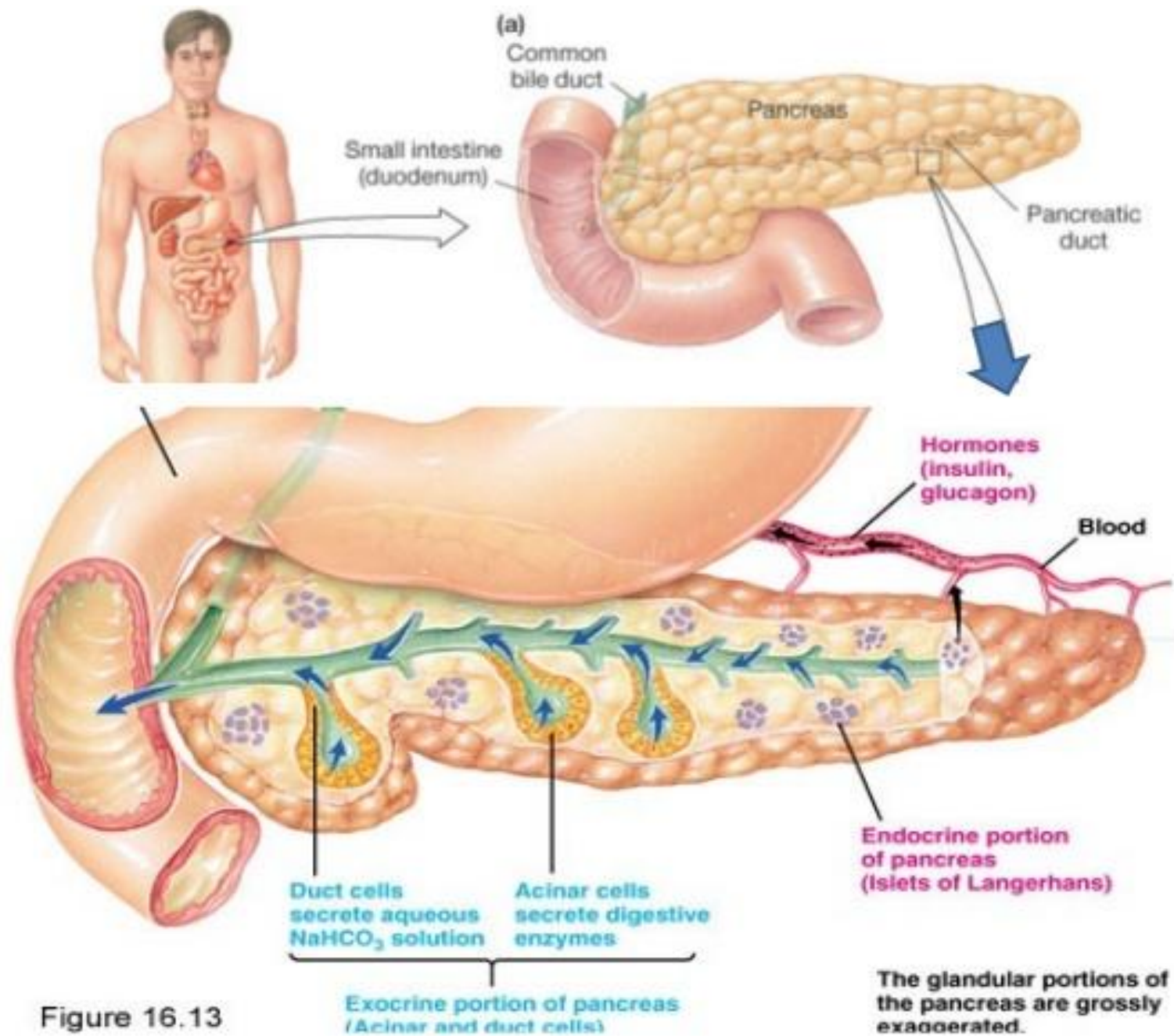
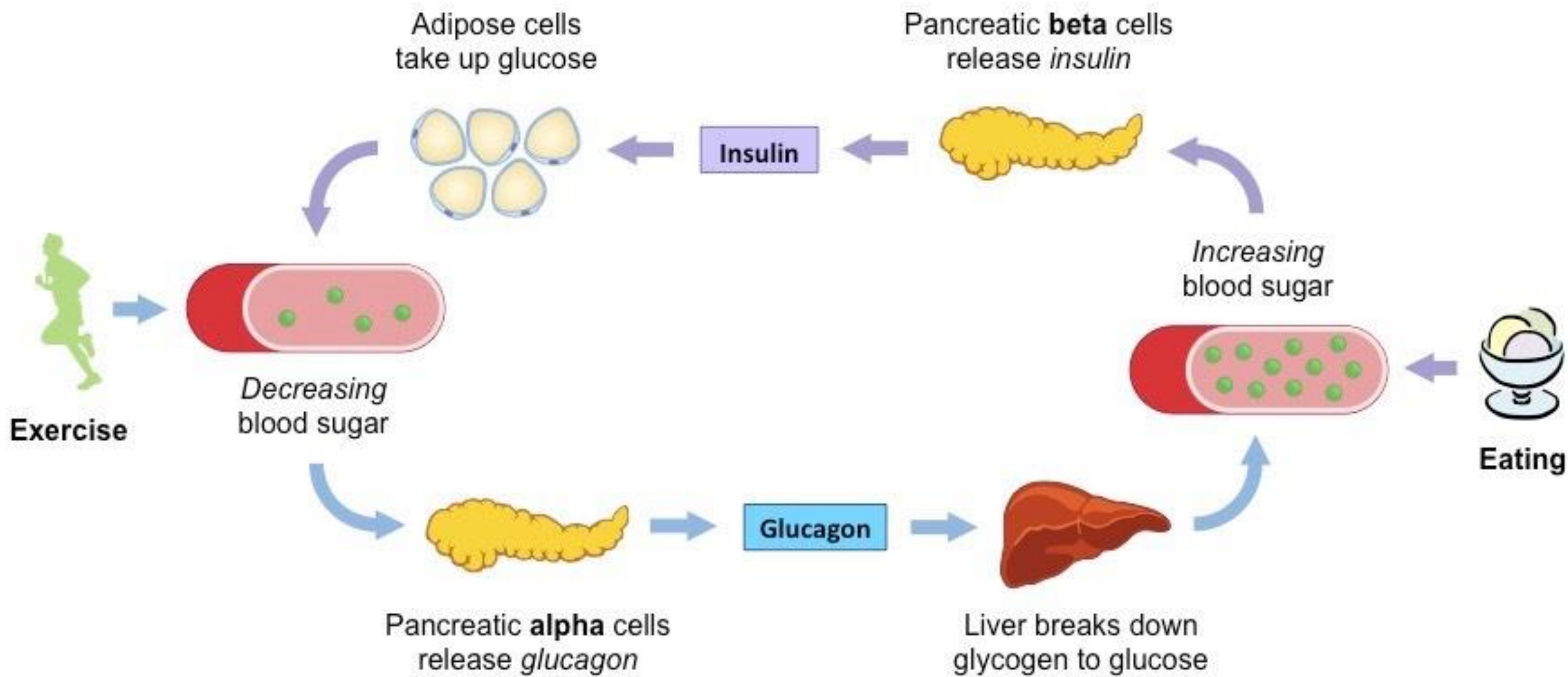
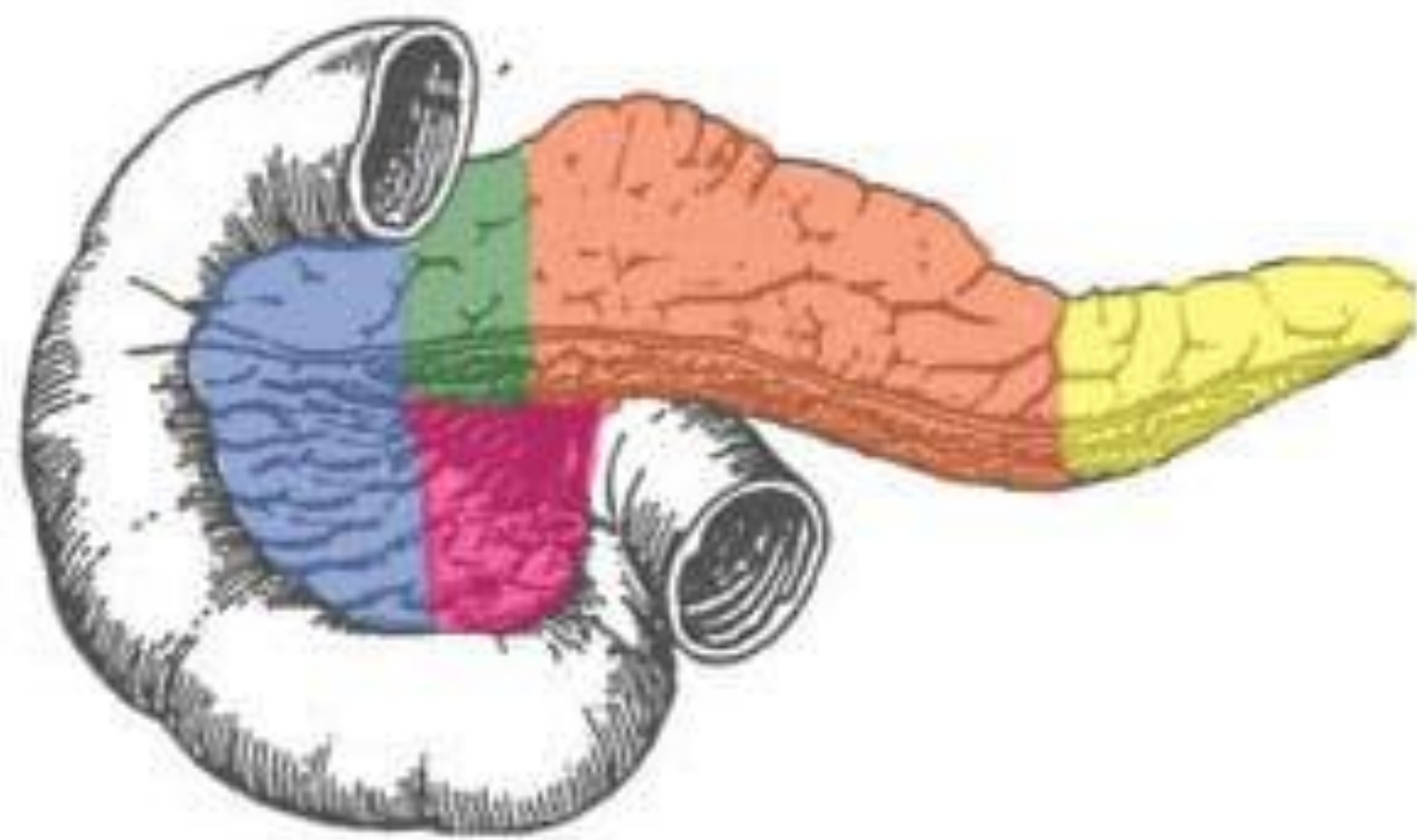
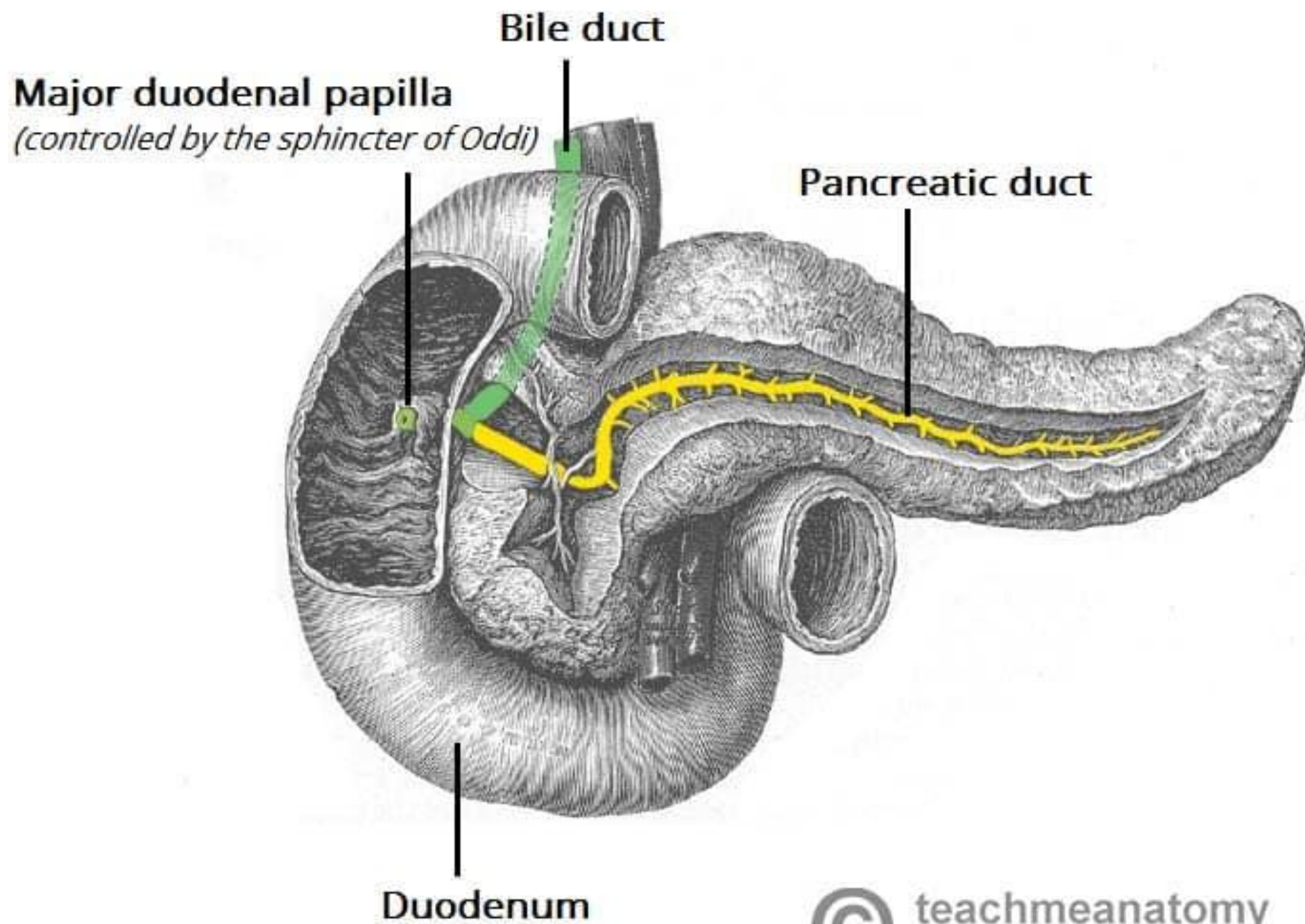


Figure 16.13



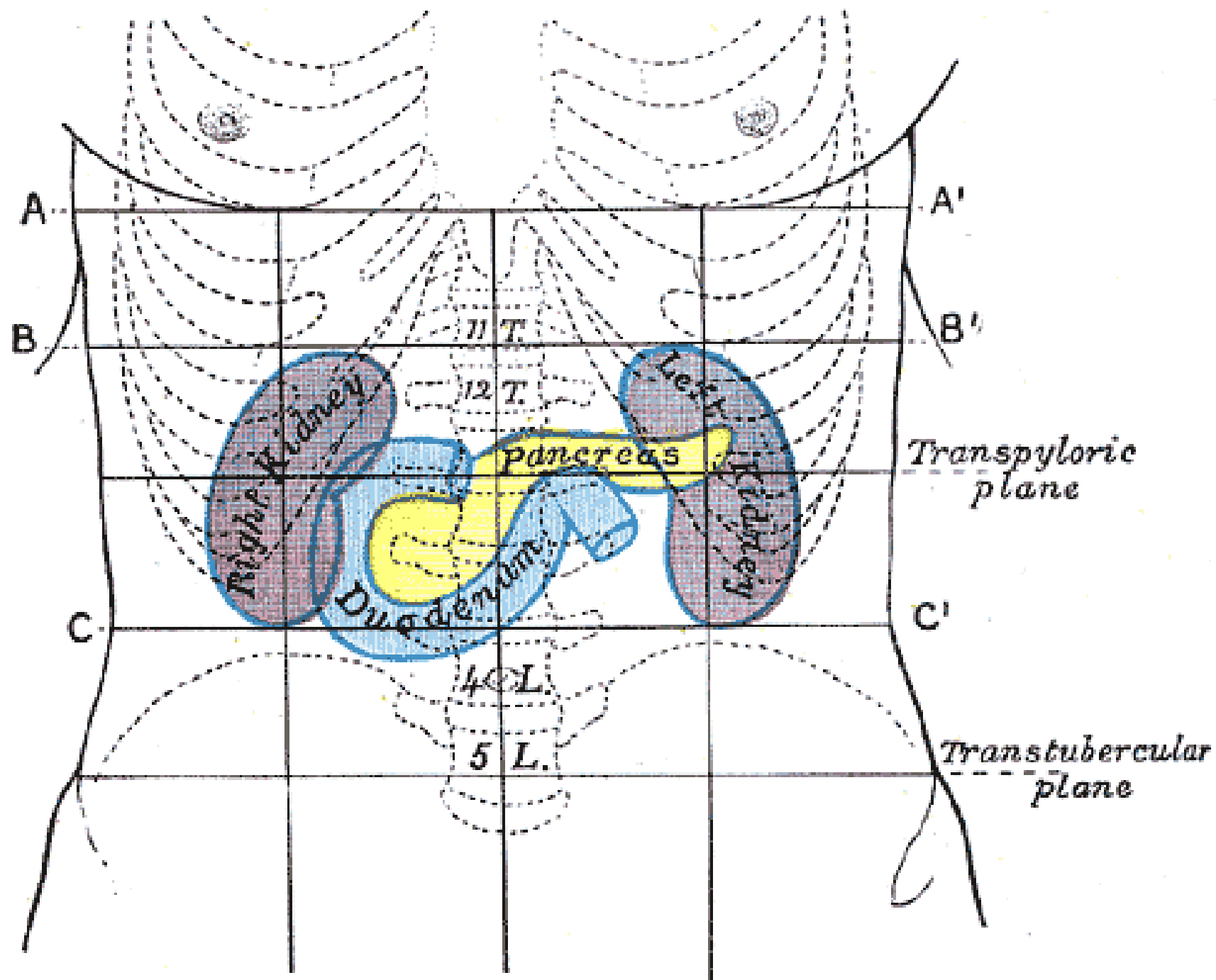


-  **Uncinate process**
-  **Head**
-  **Neck**
-  **Body**
-  **Tail**

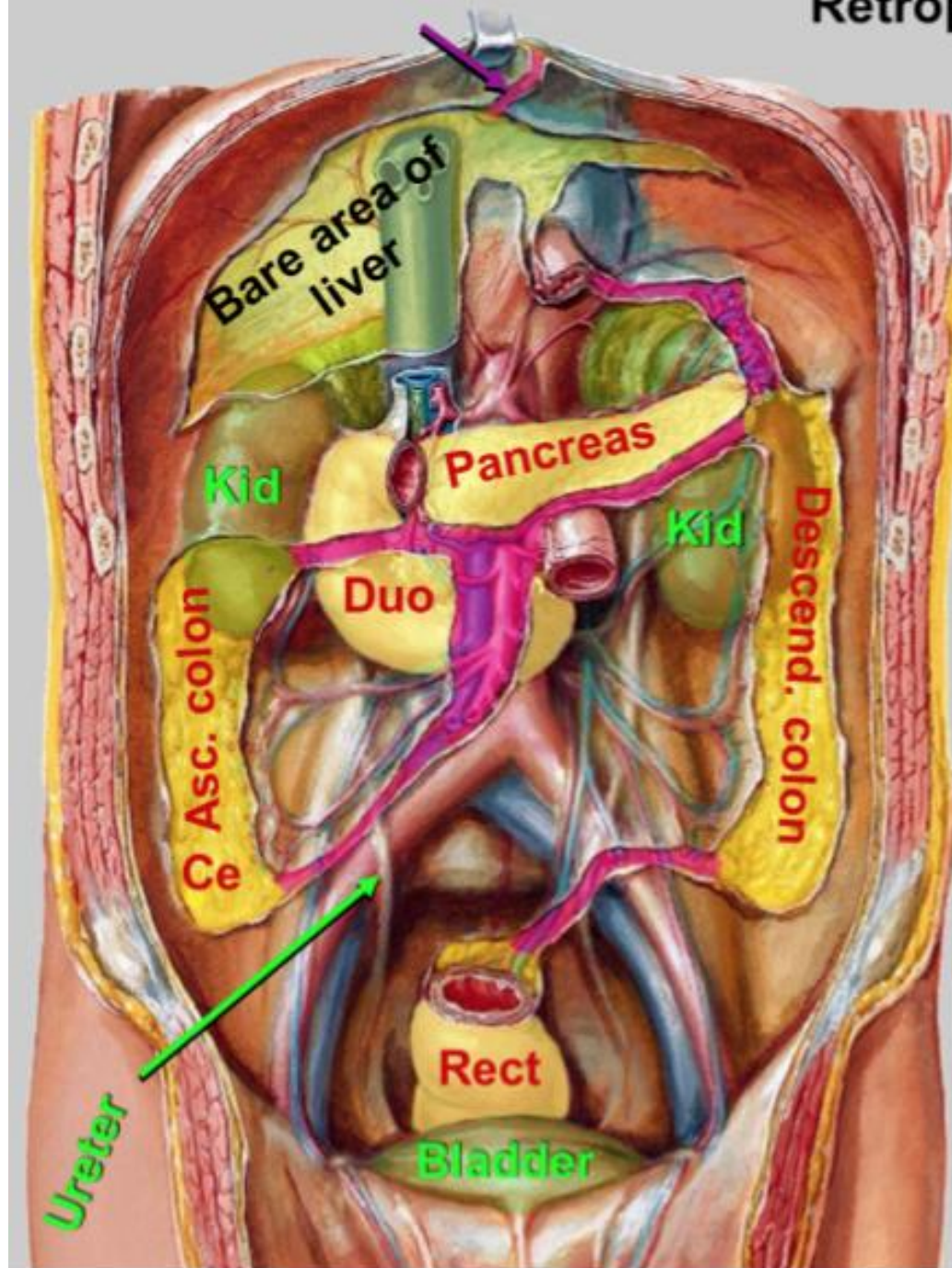


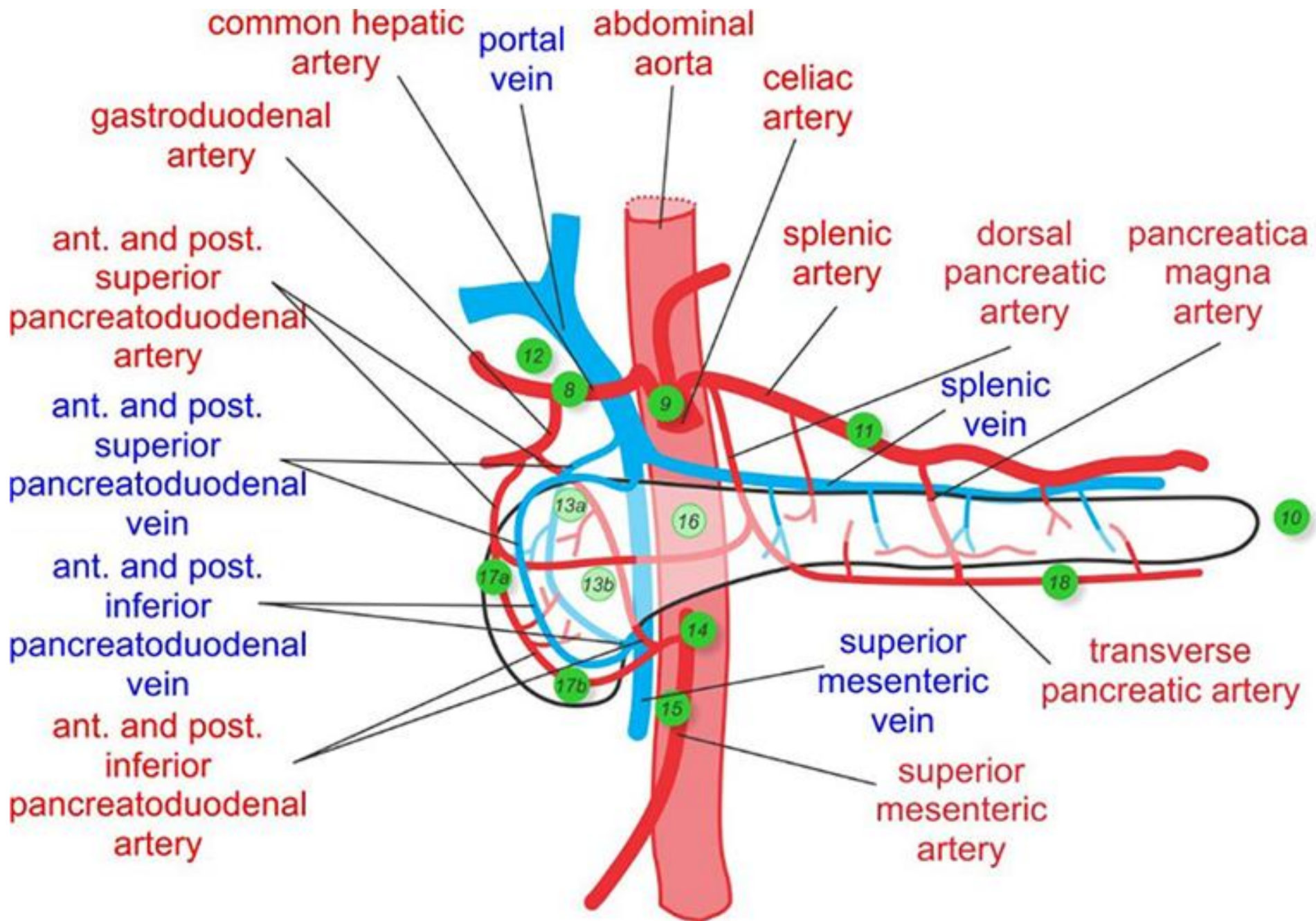
teachmeanatomy

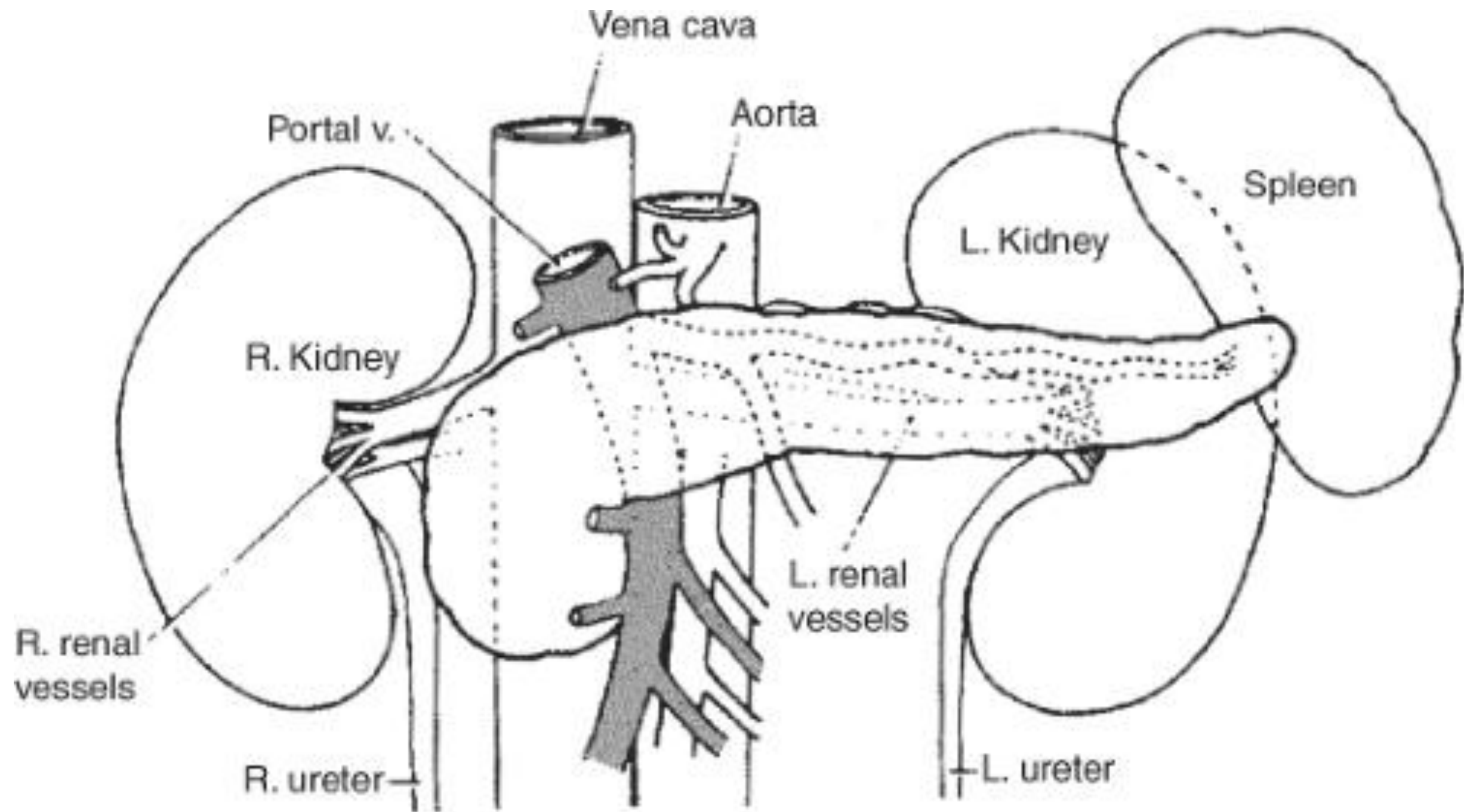
The #1 Applied Human Anatomy Site on the Web.

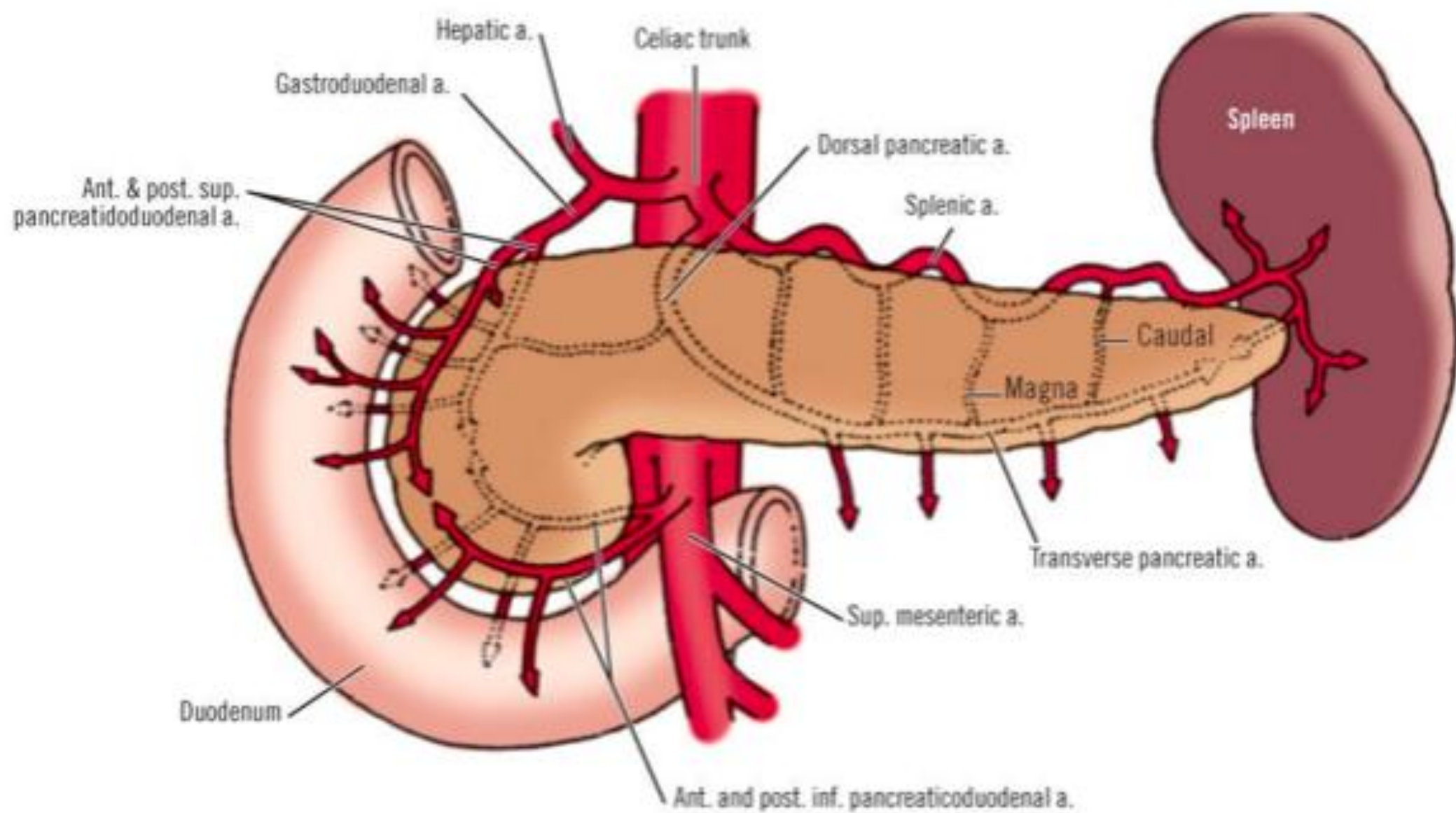


Retroperitoneal ("Fixed") Viscera



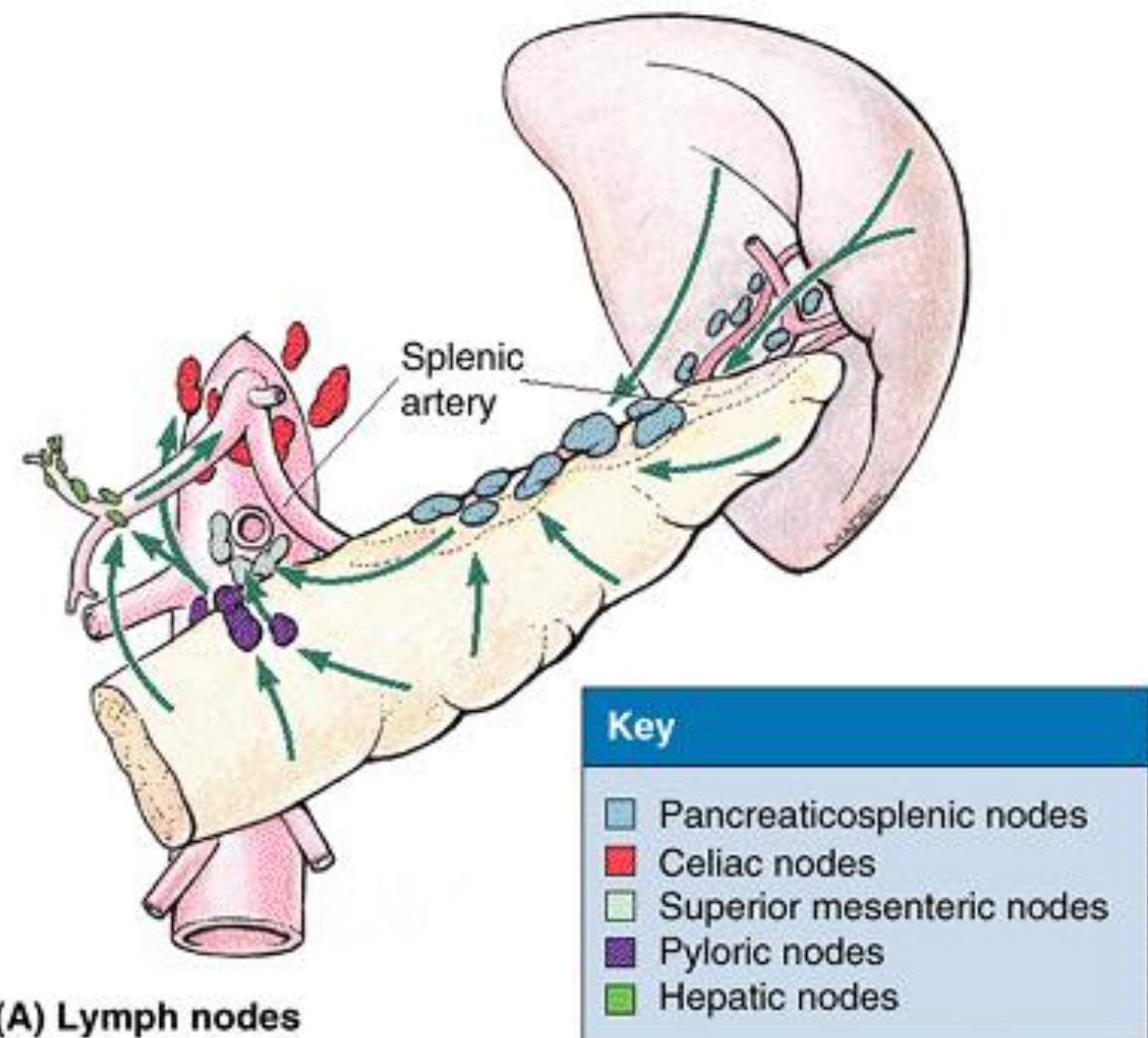




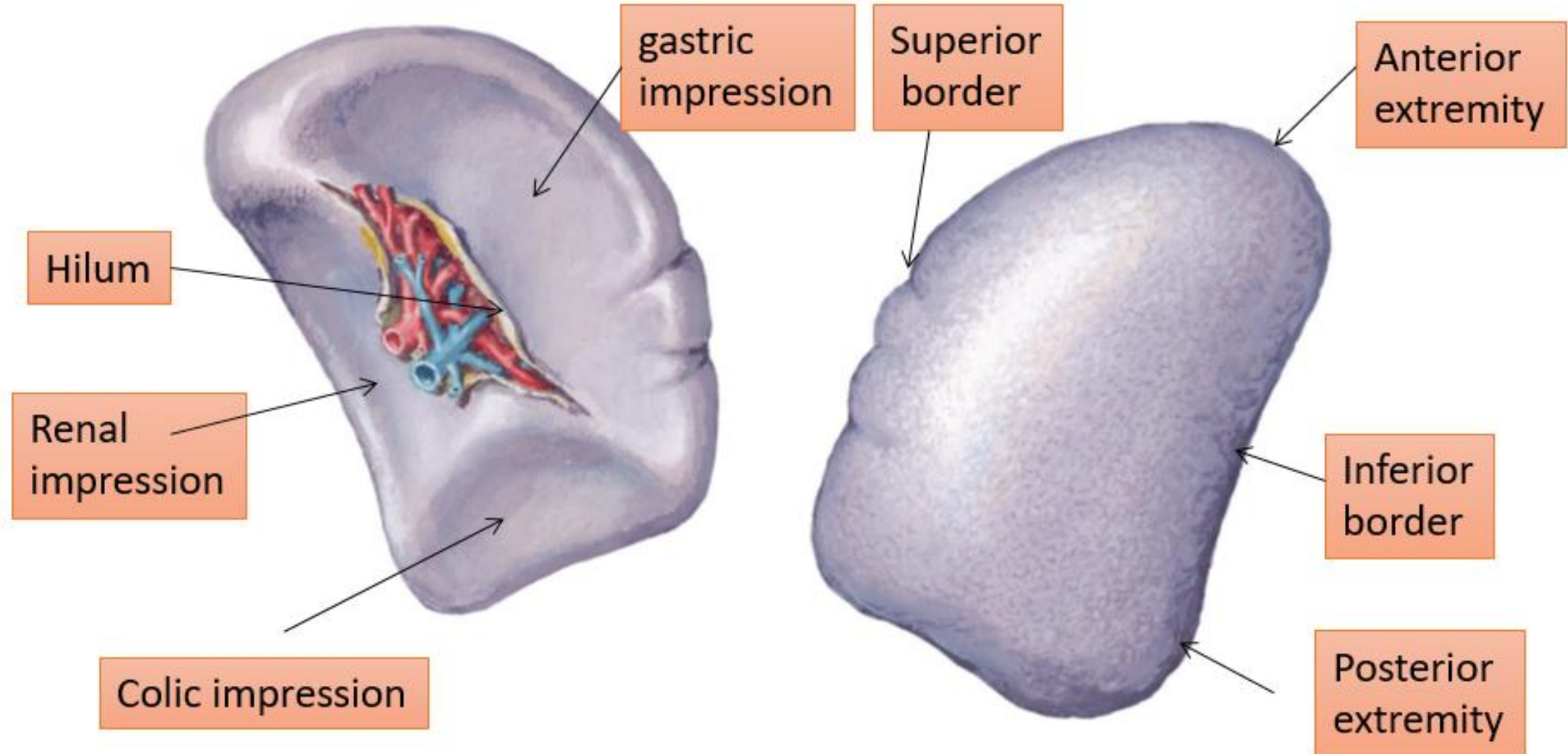


lymphatic drainage

- 1- Upper part of head → coeliac lymph nodes.
- 2- Lower part of head → superior mesenteric lymph nodes.
- 3- Rest of pancreas → pancreatico-splenic lymph nodes.



Surfaces of the spleen



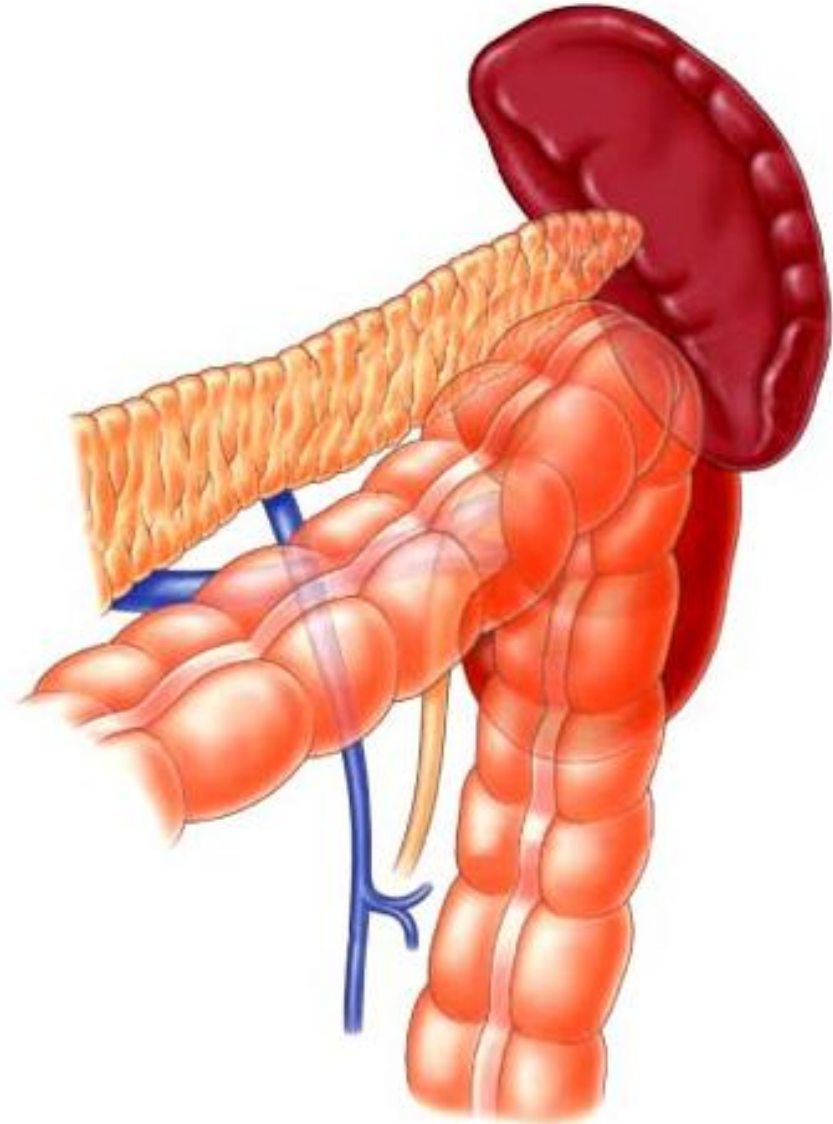
Size, shape and position

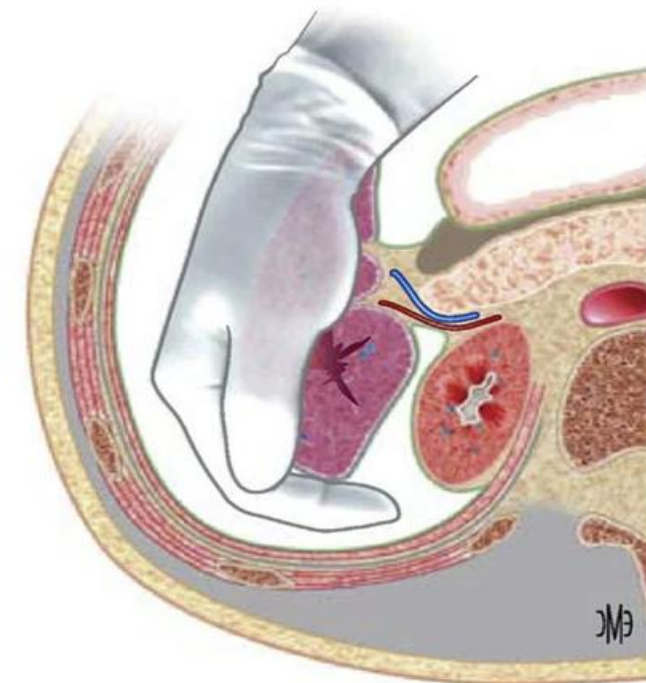
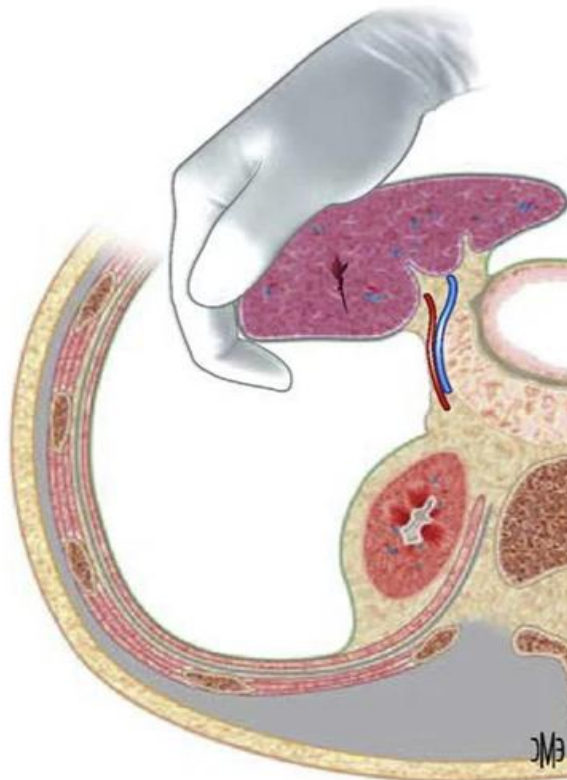
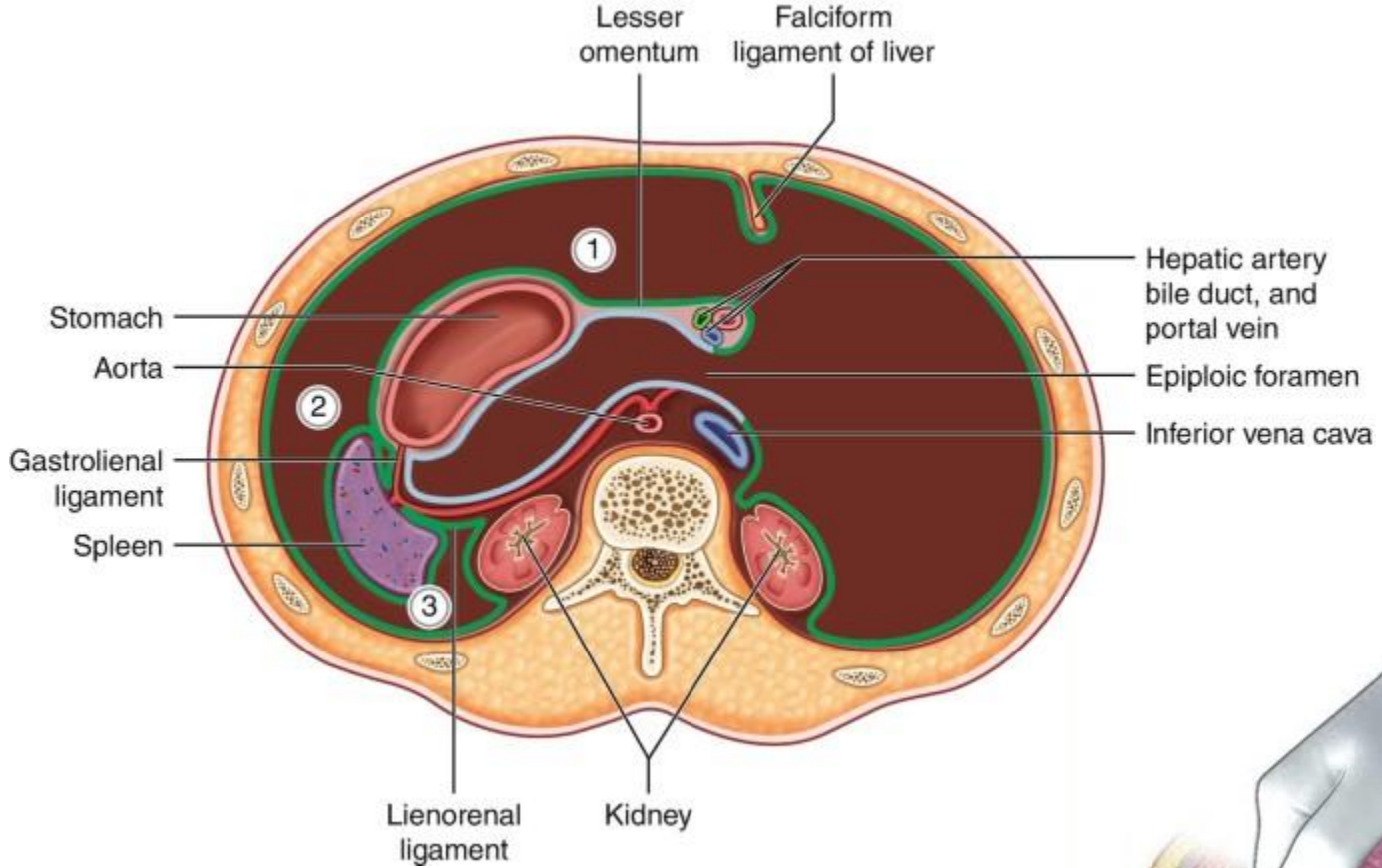
- Harris's odd numbers **1, 3, 5, 7, 9, and 11** is useful for remembering certain average dimensions of the spleen:

- The spleen measures **1 x 3 x 5** inches (2.5 x 7.5 x 12.5 cm)
- The spleen weighs **7** oz (220 g)
- The spleen relates to left ribs **9** through **11**

Relations of spleen

- **Diaphragmatic surface** - diaphragm
- **Visceral surface**
 - Anteriorly - fundus of stomach
 - Posteriorly - left suprarenal gland and kidney
 - Inferiorly - tail of pancreas and left colic flexure





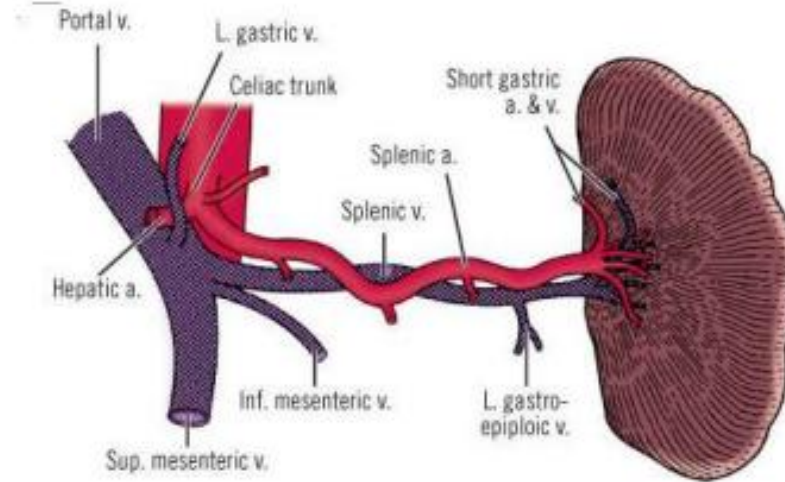
Artery:

- Splenic Artery.
- Very Tortuous
- Enters the hilum between layers of the splenorenal ligament
- Divides into 5-6 branches which enters the spleen

Vein:

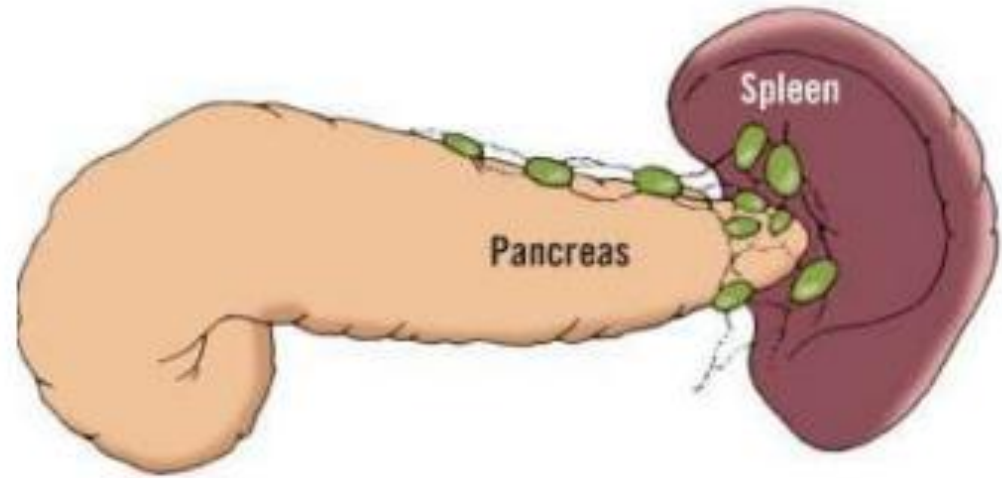
- Usually posterior to the artery
- Empties into the portal vein
- Receives also the inferior mesenteric vein
- Not as tortuous as the artery

Blood supply



Lymphatic drainage

The splenic lymphatic vessels leave the lymph nodes in the splenic hilum and pass along the splenic vessels to the pancreaticosplenic lymph nodes



THANKS



Trust only on science....

Parts of pancreas

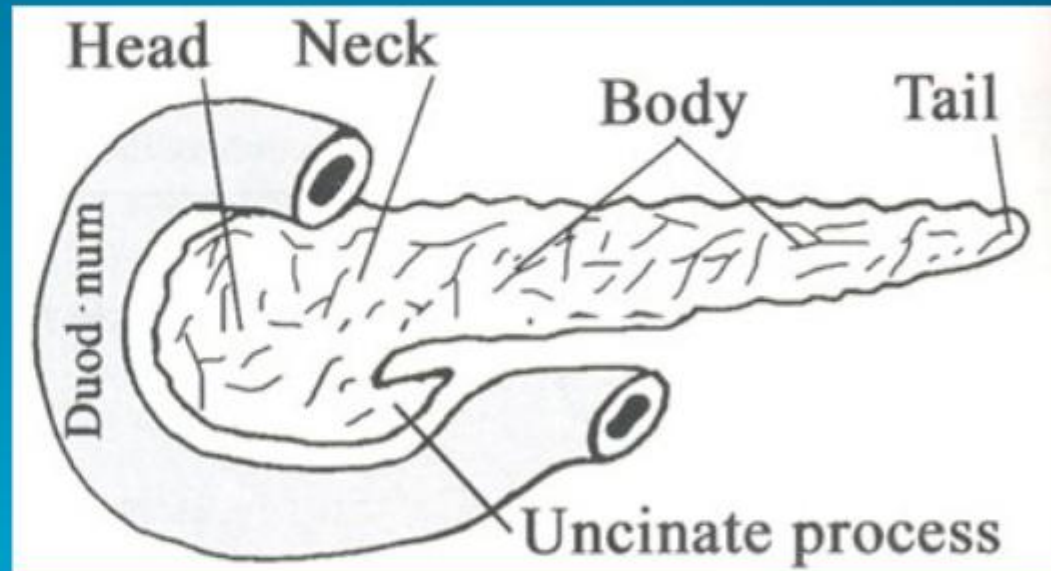
1- Head

2- neck

3- Body

4- Tail

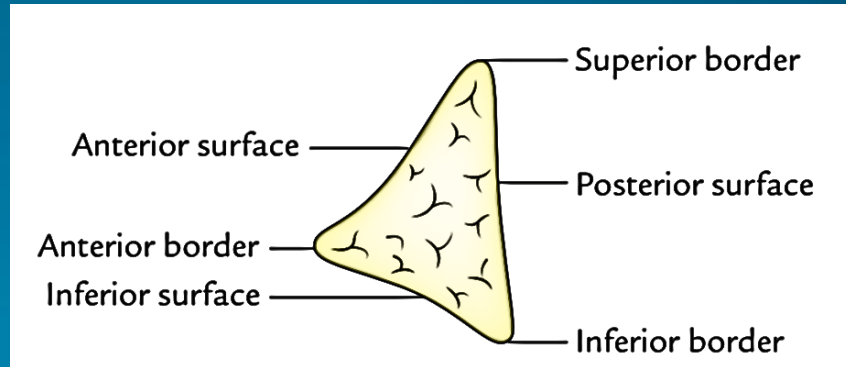
- The lower part of the head forms a projection called uncinata process.



• Borders:

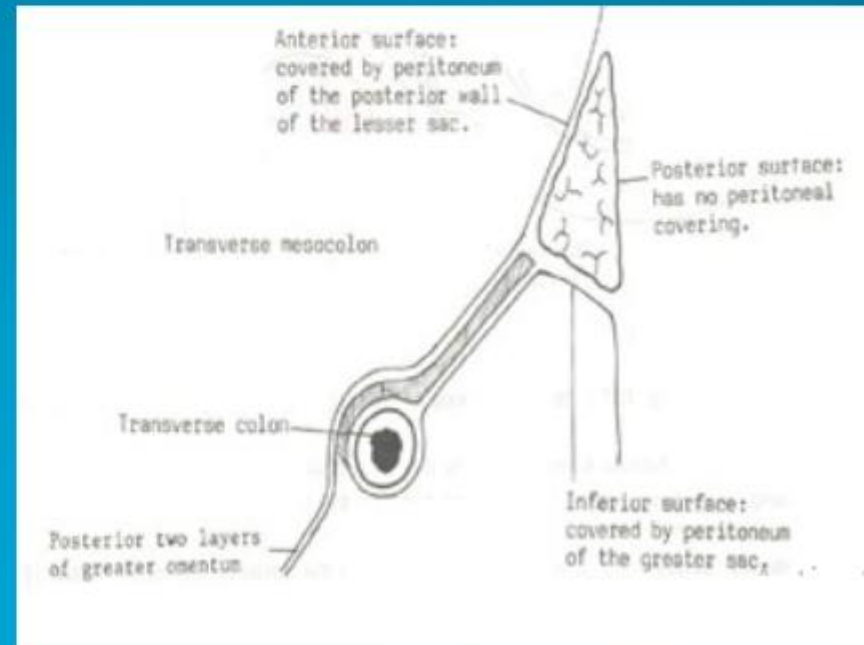
1- Superior border:

It is related to splenic artery.



2- Anterior border:

It gives attachment to transverse mesocolon and greater omentum.



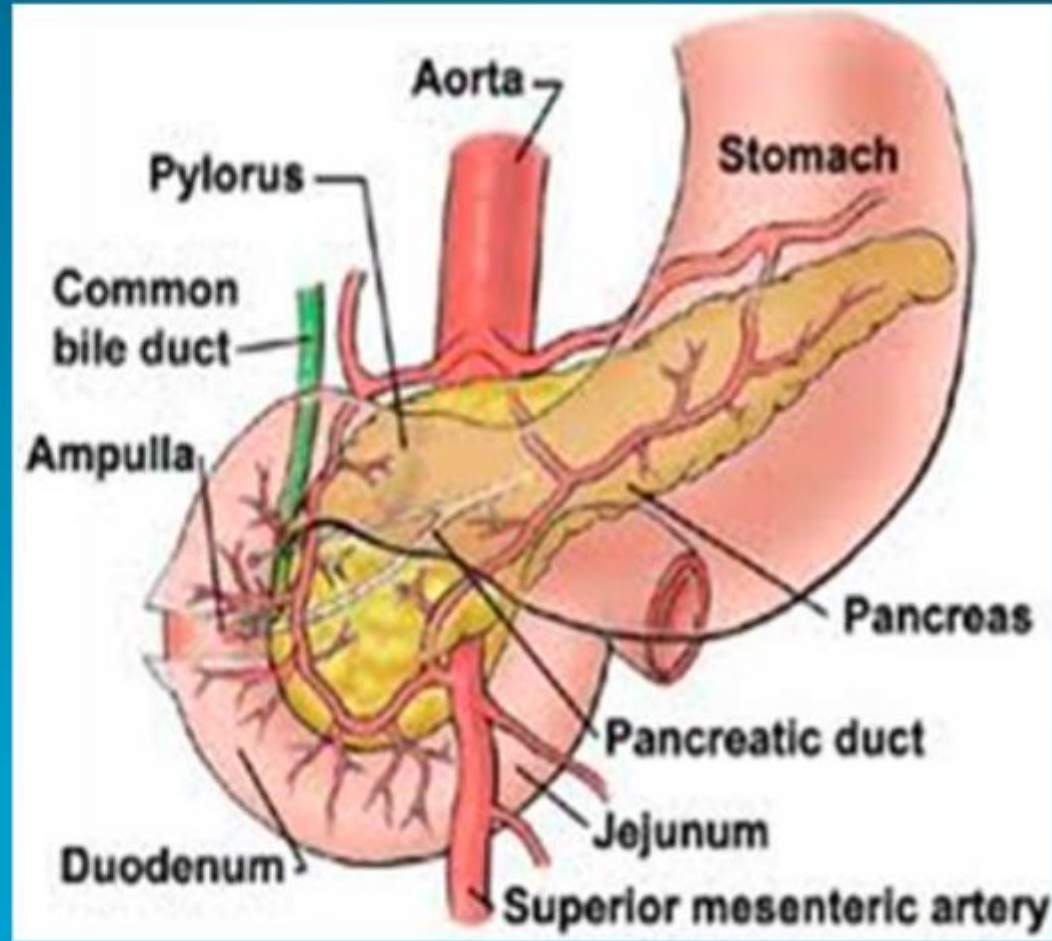
3- Inferior border:

It separates the inferior from the posterior surfaces.

Relations of pancreas

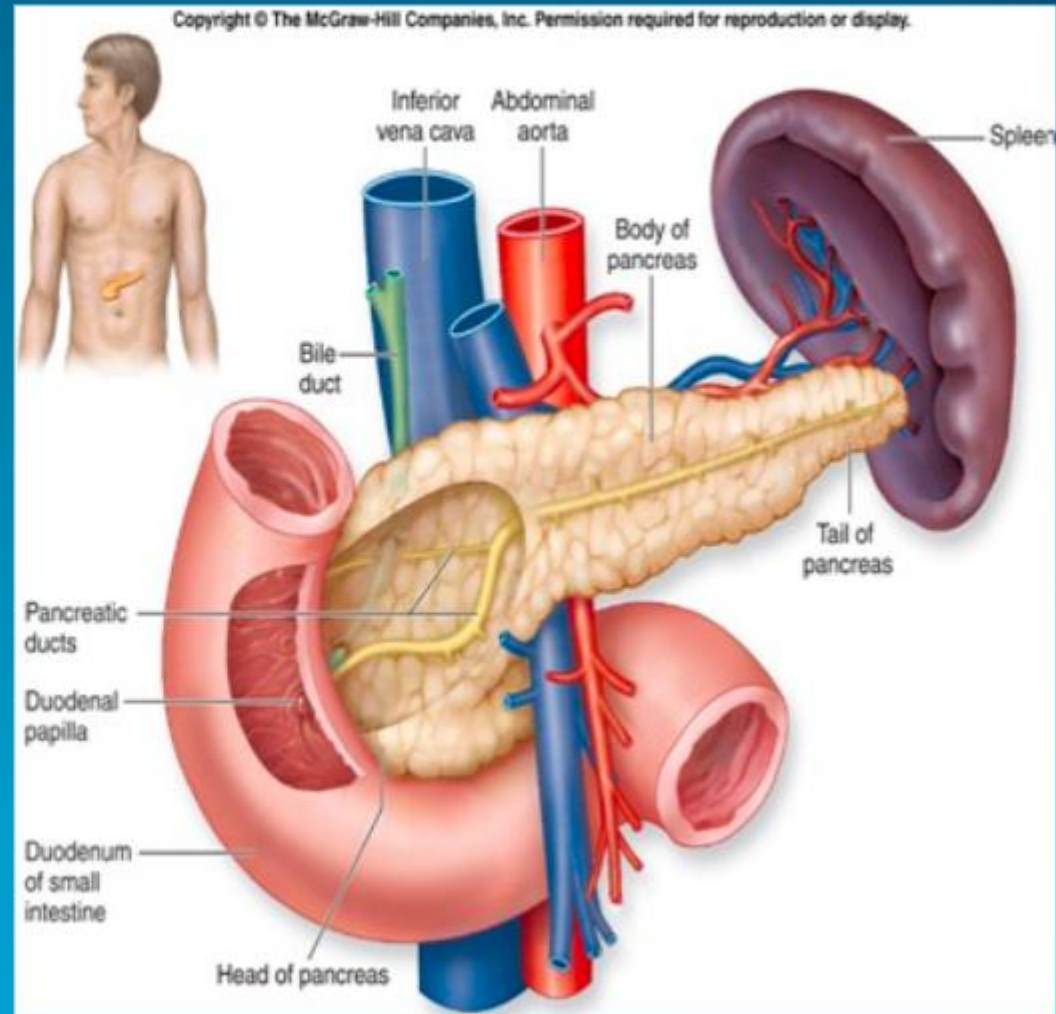
Head of pancreas:

- It lies in the concavity of the duodenum.
- It is related to the 1st part of duodenum superiorly, 2nd part on the right side (separated from it by sup. & inf. pancreatico - duodenal arteries) and 3rd part inferiorly.



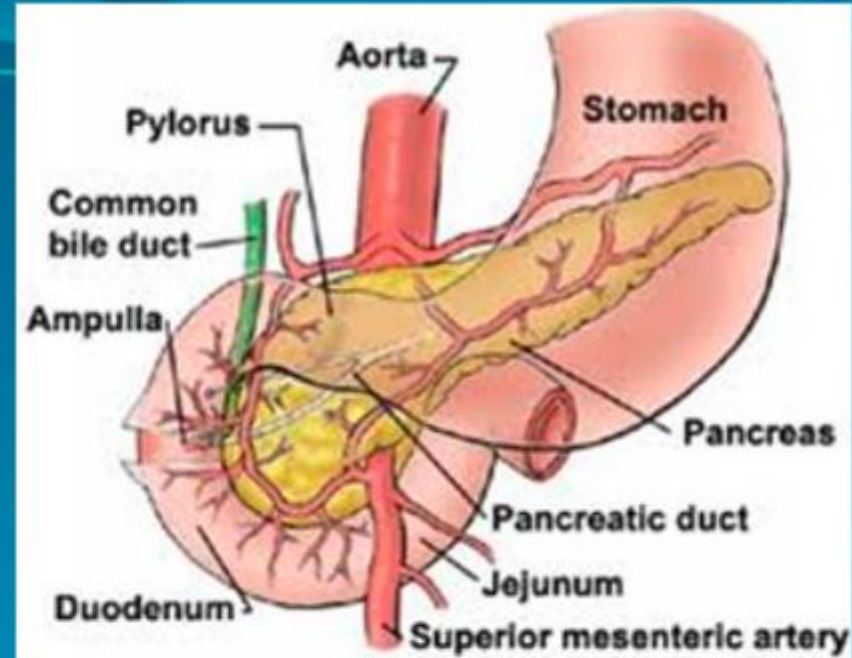
Head of Pancreas

- Anteriorly , it is related to transverse colon.
- Posteriorly , it is related to IVC, renal veins and common bile duct.
- Uncinate process lies between abd. aorta and superior mesenteric vessels.



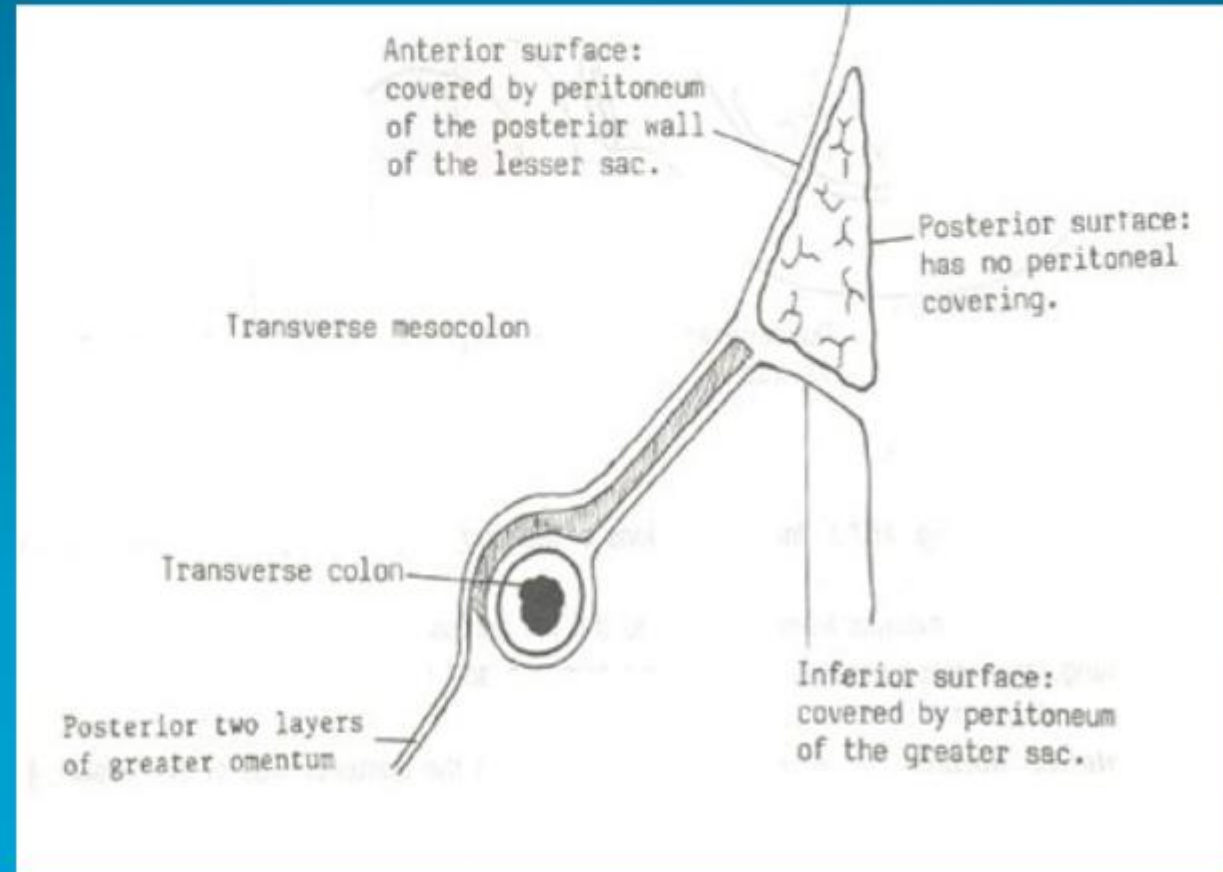
Neck of pancreas:

- **Anteriorly:**
it is related to gastro-duodenal junction.
- **Posteriorly:**
it is related to the formation of portal vein from splenic and superior mesenteric veins.



Body:

- It has three surfaces (anterior, posterior and inferior) and three borders (anterior, superior and inferior).



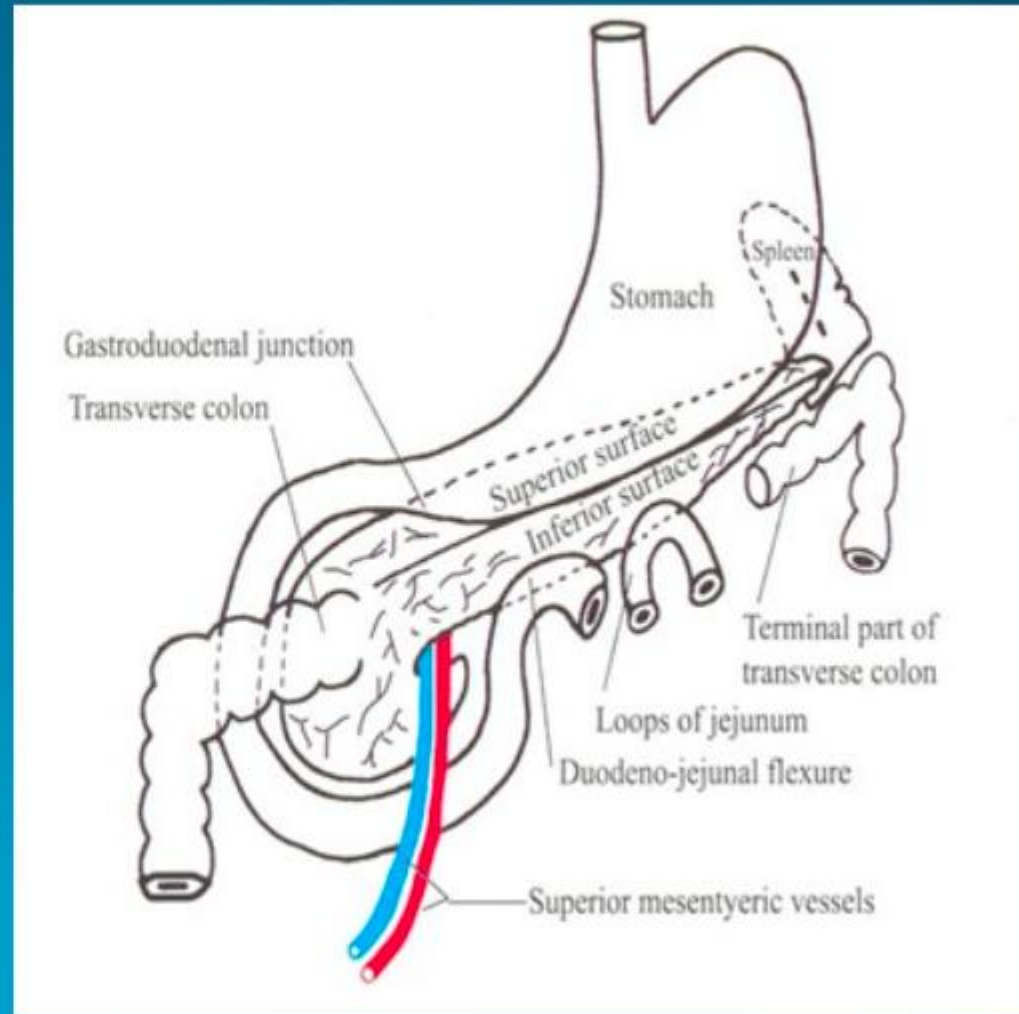
- Surfaces of body of Pancreas:

- 1- Anterior surface:

Related to stomach , separated from it by the lesser sac.

- 2- Inferior surface:

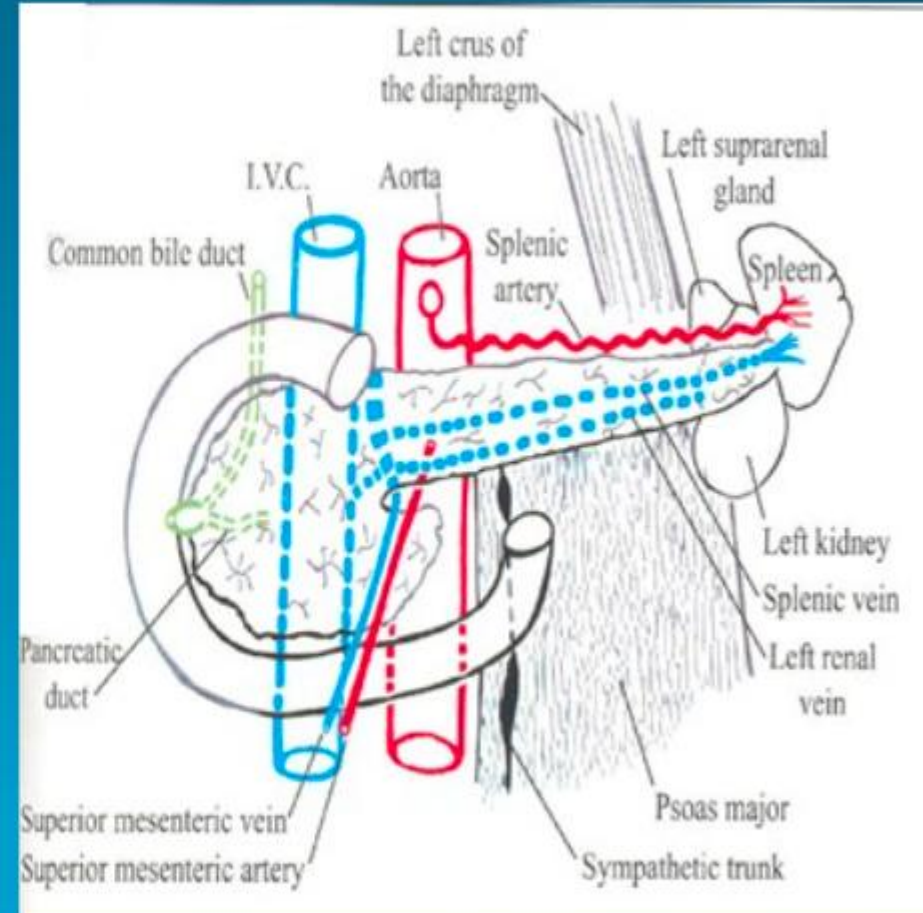
Related to duodeno-jejunal flexure, loops of ileum and end of transverse colon (from right to left)



Body of pancreas:

3- Posterior surface:

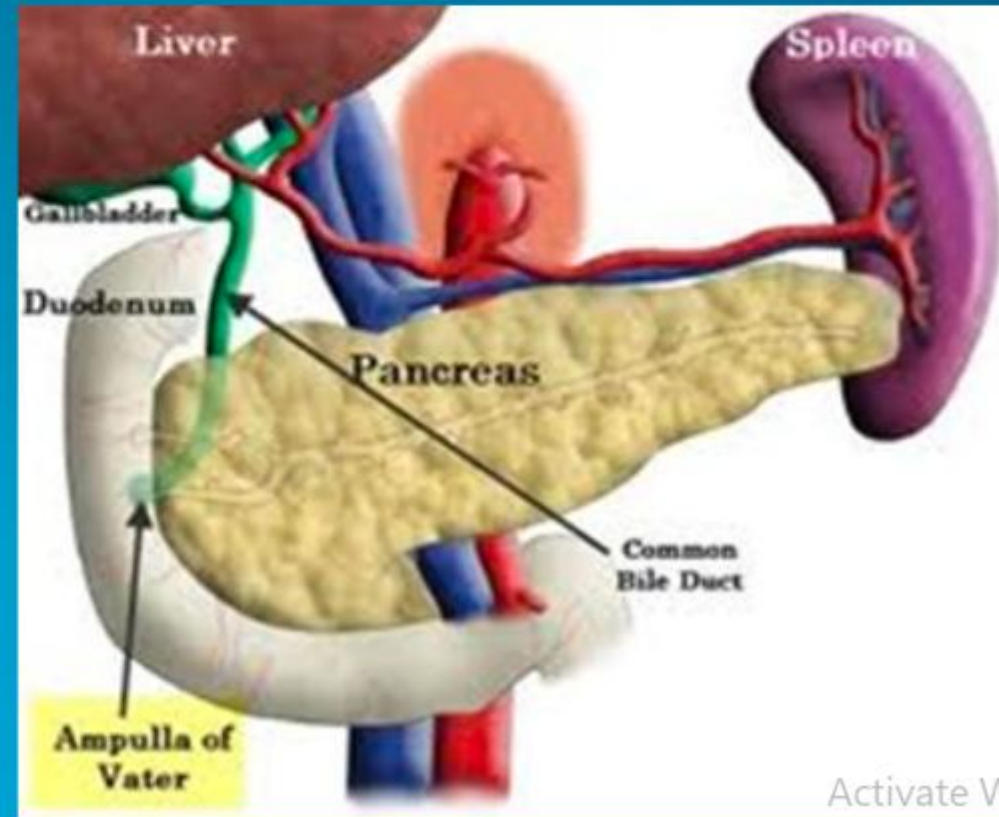
- It is related to posterior abd. wall:
- 1-Aorta and origin of sup. Mesenteric artery.
 - 2-Splenic and left renal vein.
 - 3-Left psoas major.
 - 4- Left crus of diaphragm.
 - 5- Left kidney.
 - 6- Left supra renal gland.
 - 7- Left sympathetic chain.



Tail of pancreas:

1- It is related to the visceral surface of spleen near its hilum.

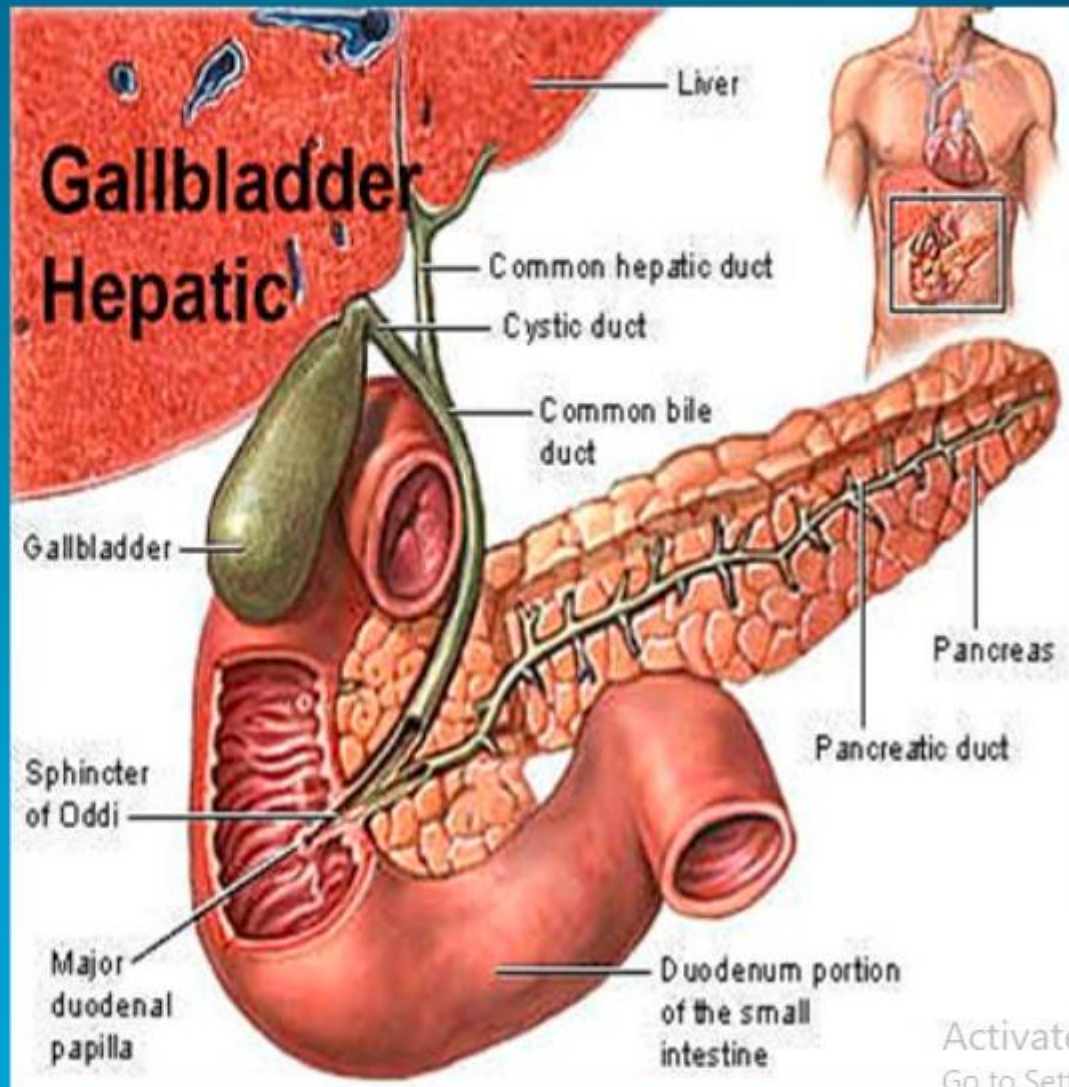
2- It reaches the hilum via the lienorenal ligament.



Ducts of pancreas

1- Main pancreatic duct:

- It drains the upper part of the head, all the body and tail of pancreas.
- It runs from the tail to the head, then it unite with common bile duct to form ampulla of Vater which opens in the 2nd part of duodenum (postero-medial).



2- Accessory pancreatic duct:

It drains the uncinate process and lower part of head.

It opens in the 2nd part of duodenum above the ampulla of Vater.

